Narrated PowerPoint Presentations

1. **Overview & Principles of Endocrine System**
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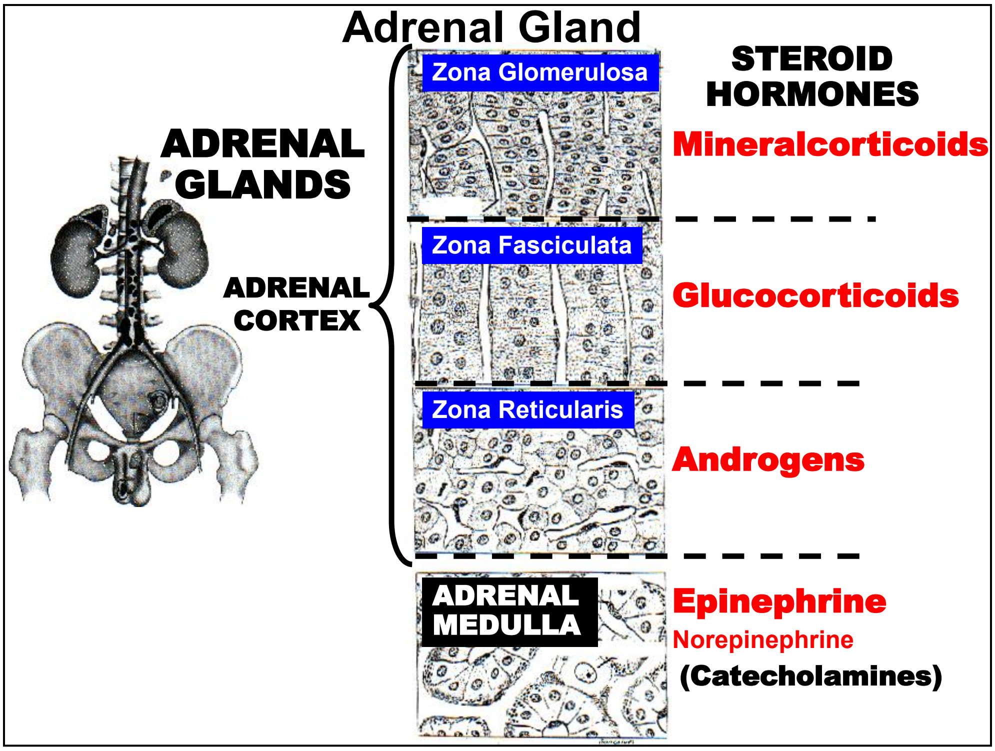
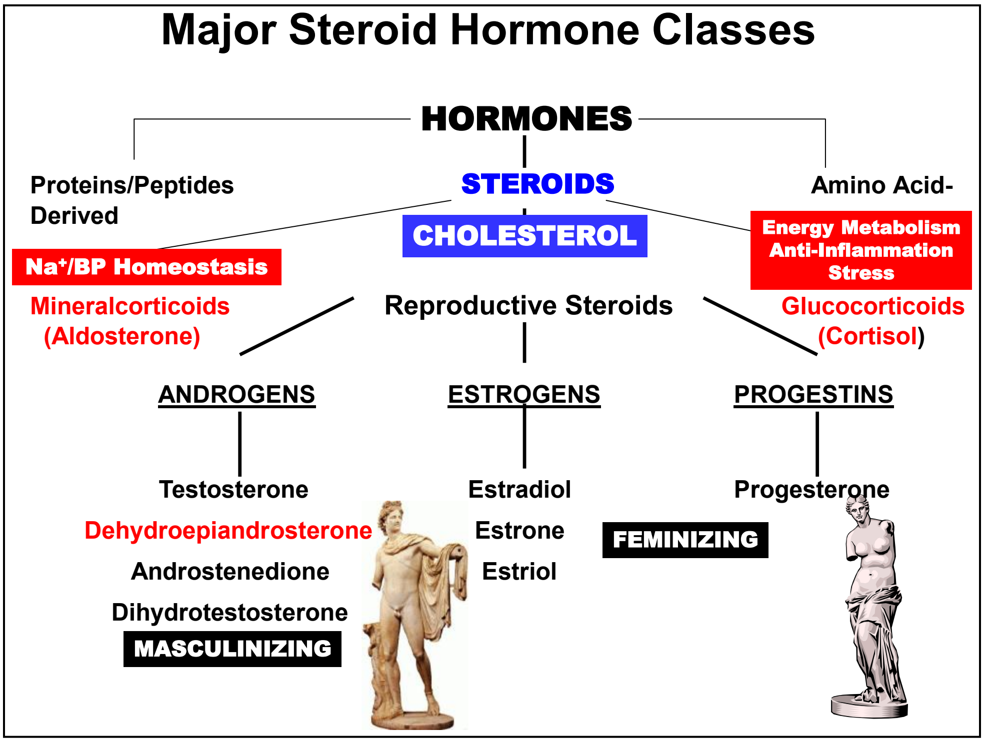


(*left*) The adrenal glands are situated bilaterally above the kidneys and are histologically and functionally divided into a larger (75% mass) outer cortex (**Adrenal Cortex**) region and an inner medulla (**Adrenal Medulla**).

(*right*) The Adrenal Medulla (*bottom region*) is considered a modified sympathetic ganglion that secretes **Catecholamines** (mainly **Epinephrine**) into blood in response to preganglionic sympathetic stimulation. The mesodermally derived Adrenal Cortex (*top 3 regions*) produces a class of lipid-soluble hormones derived from Cholesterol called **STEROIDS**.

The cortex is subdivided into 3 zones that secrete the 3 major classes of cortical hormones: The Zona Glomerulosa (*upper zone*), exclusively produces **Mineralcorticoid** hormones. The Zona Fasciculata (*middle zone*) primarily produces **Glucocorticoids** and the Zona Reticularis (*bottom zone*) almost exclusively produces **Androgens**.



Review of the various classes of endocrine hormones, including steroid hormones.

Note that **Reproductive Steroids** are a general class of **Steroid Hormones**. There are three major subclasses of Reproductive Steroid hormones based on their biochemical structure and, in part, on their distinct general physiological effects:

Androgens are the predominant subclass of reproductive hormone in males. The most abundant androgen hormone secreted in males is **Testosterone**. Other important androgens of physiological importance are listed as well. Physiologically, androgens exert a "Masculinizing" effect. That is, androgens stimulate processes that result in a phenotype generally more typical of males and, in particular, primary and secondary sexual characteristics.

Estrogens are one of the two predominant subclasses of reproductive hormones in females. The most abundant Estrogen hormone secreted in females is **Estradiol**. Other important estrogens of physiological importance are listed as well. Physiologically, estrogens exert a "Feminizing" effect. That is, estrogens stimulate processes that result in a phenotype generally more typical of females and, in particular, primary and secondary sexual characteristics.

Progestins are the other predominant subclass of reproductive hormones in females. The most abundant Progestin secreted in females is **Progesterone**. Physiologically, progesterone is essential for certain female reproductive processes described later.

The role of Androgens in male reproduction is discussed next.

Steroid Hormone Properties

- **Classification**: **Steroid (Cholesterol)**
- **Biosynthesis**: **Adrenal Cortex-Gonads-Placenta**
- **Storage**: **No**
- **Secretion**: **Passive Diffusion**
- **Transport**: **Bound (50-90%)**
- **Metabolism**: **Biochemical Inactivation** (Half-Life: 20-90 min)
- **Mechanism**: **Intracellular Membrane Receptor**
- **Actions**: **Energy Met., Homeostasis, Growth & Dev., Reproduction**
- **Regulation**: **Cortisol: Hypothalamic-Ant. Pit. (CRH/ACTH)**
  Aldosterone: Renin-Angiotensin
- **Disorders**: **Excess: Cortisol: Cushing's**
  Deficiency: Cortisol: Addison's

Steroid Hormone endocrine properties

Highlights are summarized above.

Glucocorticoids

Glucocorticoid Actions

CATABOLIC

Carbohydrate → Glucose
Protein → Amino Acids
Fat → FFA

Energy Metabolism

Anti-Inflammation

Anti-Stress

Although named for their critical role in maintaining carbohydrate reserves, **Glucocorticoids** produce a number of diverse actions in virtually every tissue of the body.

Most can be related to three major processes:

Energy Metabolism (Net Catabolic Effect)

Anti-Inflammation

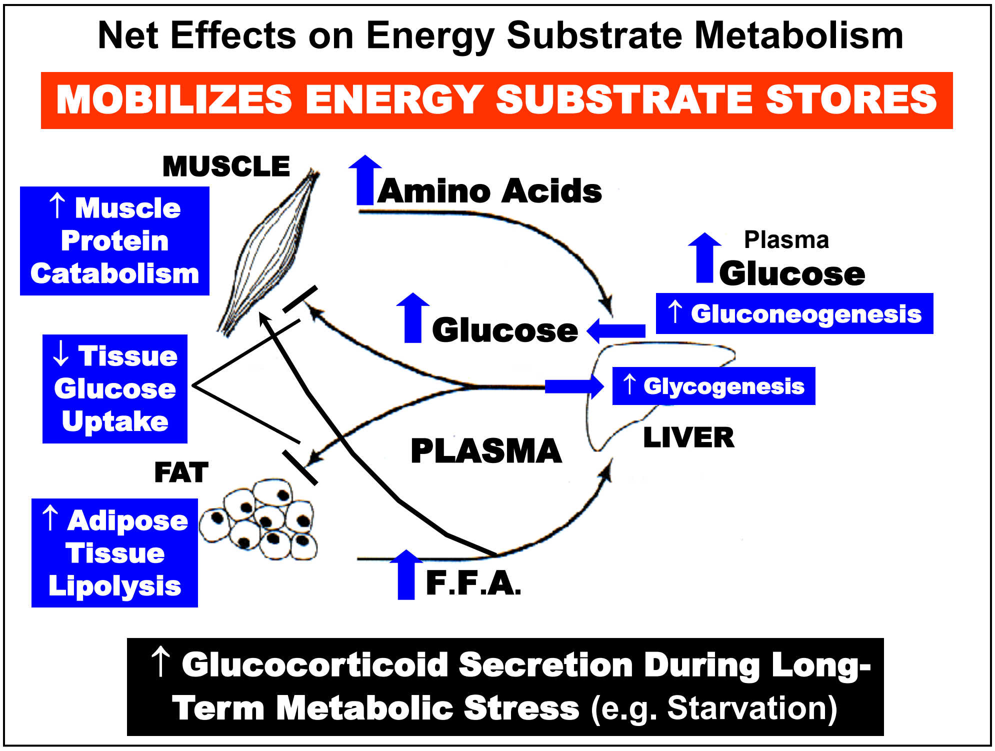
Anti-Stress

Cortisol is the adrenal steroid responsible for the vast majority (≈95%) of the physiological activities associated with glucocorticoid secretion in humans. The remaining activity (5%) is due to **Corticosterone** and **Cortisone**.

Glucocorticoid Actions on Organ Systems

TISSUE	EFFECT
CENTRAL NERVOUS SYSTEM	Taste, Hearing & Smell Acuity Regulates CRH Secretion: ↑ Addison's; ↓ 1° Cushing's Syndrome ↓ ADH Secretion
CARDIOVASCULAR SYSTEM	Maintain Catecholamine Sensitivity ↑ Vasoconstrictor Agent Sensitivity Maintain Microcirculation
GI TRACT	↑ Gastric Acid Secretion ↓ Gastric Mucosal Cell Proliferation
LIVER	↑ Gluconeogenesis; Glycogen & Protein Synth.
PULMONARY SYSTEM	↑ Surfactant Production & Fetal Maturation
PITUITARY GLANDS	↓ ACTH Secretion (Acute) & Synthesis (Chronic)
RENAL SYSTEM	↑ GFR Dilute Urine Excretion
SKELETAL SYSTEM	↑ Resorption ↓ Formation
MUSCULAR SYSTEM	↓ Fatigue (2° to CV Actions) ↑ Protein Catabolism ↓ Protein Synthesis ↓ Glucose Oxidation ↓ Insulin Sensitivity
IMMUNE SYSTEM	↓ Mass of Thymus & Lymph Nodes ↓ White Blood Cell Concentrations ↓ Cellular Immunity
CONNECTIVE TISSUE	↓ Activity of Fibroblasts; ↓ Collagen Synthesis

Although named for their critical role in maintaining carbohydrate reserves, **Glucocorticoids** produce a number of diverse actions in virtually every tissue of the body. These affects are summarized above for the major organ systems



The metabolic effects of glucocorticoids on energy substrate stores and their net effect on plasma levels of the free substrate are summarized above. Glucocorticoid secretion typically increases in response to a Metabolic Stress Conditions such as **Starvation**. During such conditions, glucocorticoids also serve what has been termed a Permissive role (see effects on Liver) that facilitates the actions of other hormones also secreted during the same response (e.g. Catecholamines & Glucagon). Most glucocorticoid actions on energy substrates serve to **Mobilize Energy Stores into the Blood**. Some the effects described below will make more sense in this context.

Muscle Metabolism (*top left*). Cortisol **Increases Muscle Protein Catabolism** (↑ Protein Degradation), which results in ↑ **Amino Acid** levels in Plasma. This action increases amino acid availability in parallel with **Increased Liver Gluconeogenesis** activity (*see below*). Together, these actions also promote ↑ **Plasma Glucose** levels.

Fat Metabolism (*bottom left*). Net effect is to ↑ **Adipose Tissue Lipoysis** (↑ Fat Breakdown), which results in ↑ **Free Fatty Acid** levels in plasma. Actions on adipose tissue promote ↑ **Plasma Glucose** levels indirectly. This Glucose "Sparing" effect (↓ Glucose Utilization) resulting from ↑ FFA Oxidation by both Liver & Muscle (*lower arrows*) causes ↓ Insulin Sensitivity of Fat & Muscle (*see above*). The net effect is to ↓ **Tissue Glucose Uptake** by Adipose & Muscle Tissues.

Liver Metabolism (*upper right*). Glucocorticoids stimulate (↑) **Gluconeogenesis** by the liver, contributing to their promotion of ↑ **Plasma Glucose** . Action is complementary to the protein catabolic actions of glucocorticoids on Muscle. Interestingly, glucocorticoids also (↑) **Glycogenesis**. This allows storage of glucose in the liver rather than muscle or as fat for later mobilization for use by the brain and other glycolysis-dependent tissues should glucose availability remain limited.

Anti-Inflammatory Actions

INFLAMMATION:
- **Redness**
- **Swelling**
- **Heat**
- **Pain**
- **Loss of Function**

GLUCOCORTICOIDS:

- **Stabilize Lysosomes**:
  - ↓ Proteases & Histamine
  (↓ Pain & Function Loss)
- **↓ Capillary Permeability**:
  - ↓ Edema & Exudation
  (↓ Swelling, Pain, Function Loss)
- **↓ Blood Flow**
  - ↑ Vasoconstriction
  - ↓ Vasodilation
  - ↓ WBC Infiltration
  (↓ Redness, Swelling, Heat, Pain)
- **↓ Immune Response**:
  - ↓ Cytokines
  - ↓ B & T Lymphocytes
  (Prevents Autoimmunity)

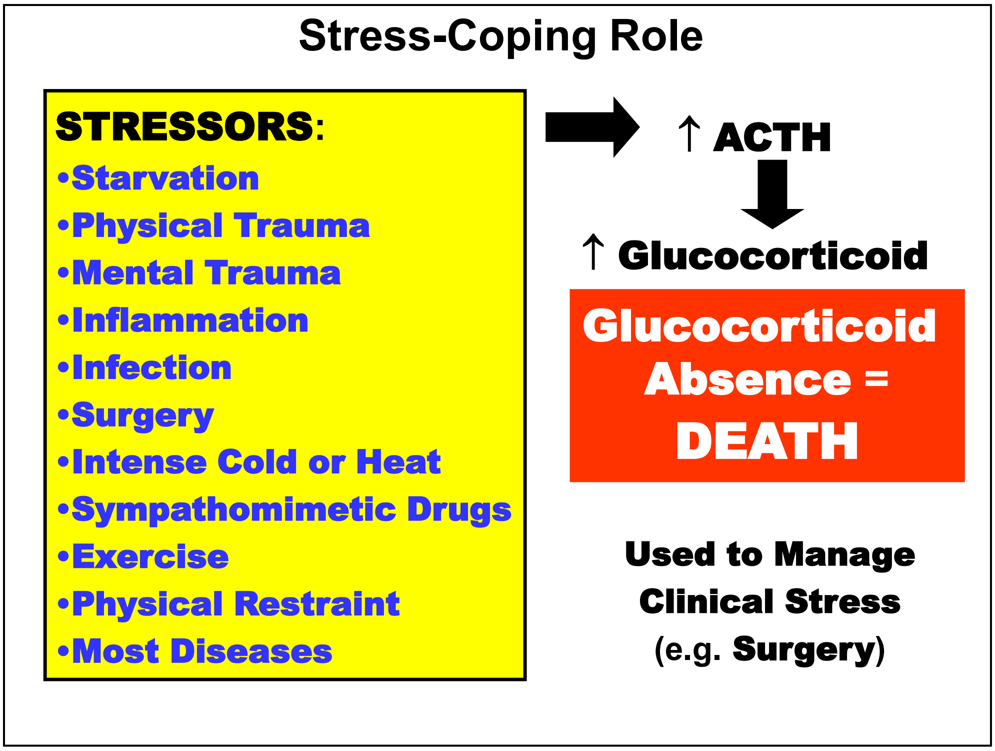
Glucocorticoids Moderates Processes Causing Inflammation

The **Inflammatory Response** is clinically characterized by the development of **Redness** and **Heat** caused by increased in blood flow to the inflamed area. **Swelling** also develops due to interstitial leakage of protein and fluid exudates caused by increased capillary permeability. **Pain** is also presented, in part, because of the increased release and delivery of chemical products (histamine, bradykinin, prostaglandins, cytokines, leukotrienes) to the site. This often involves an associated increase in white blood cell infiltration and stimulation of the immune processes. Finally, **Loss of Function** occurs and may be the direct result of the injury or the secondary consequence of the inflammatory reaction itself. Although the initial inflammatory response serves to facilitate the healing process (preventing infection, removing dead tissue & cells, delivering nutrients), an excessive response is detrimental to the tissue repair process.

Glucocorticoids act to moderate many of the inflammatory processes above that occur during trauma as well as allergic reactions and autoimmune diseases.

Clinically, glucocorticoids are commonly prescribed pharmacologically to manage pain and inflammation associated with surgery, injury and disease.

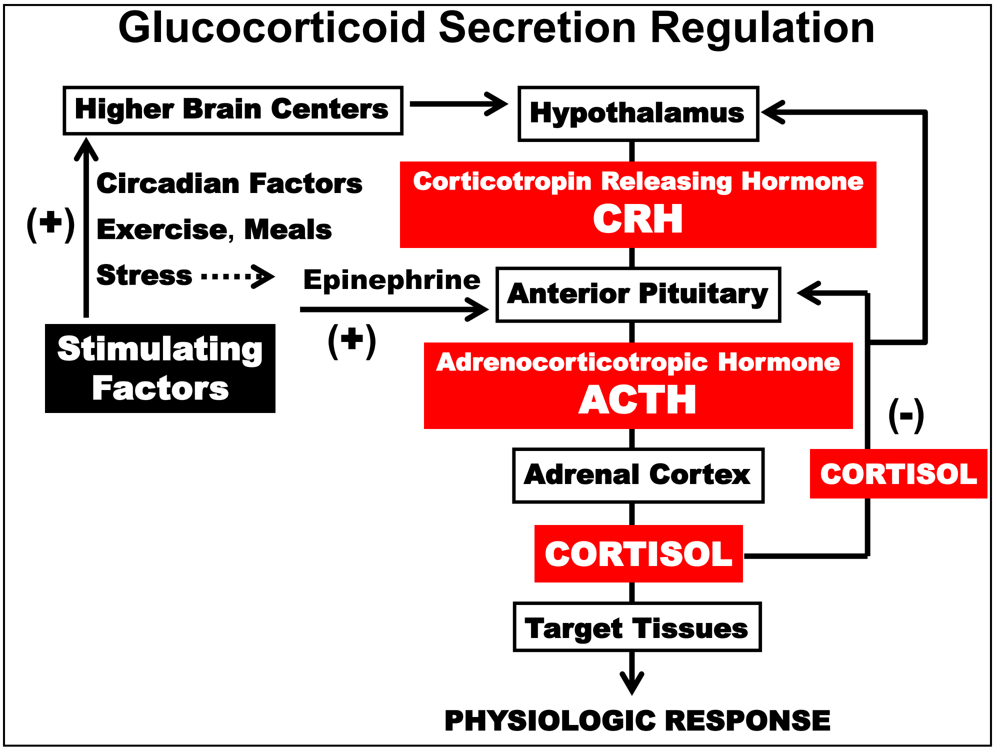
The specific anti-inflammatory actions of glucocorticoids are included in the right text box. You do NOT need to memorize the specific actions.



"Stress" was defined by the endocrinologist Hans Selye as any number of "*noxious stimuli that results in an increase in the anterior pituitary hormone ACTH*". This was based on the observation that when an animal or human is exposed to any or a variety of these so-called "Stressors" (*left*), there was a characteristic increase in **ACTH** and a consequential stimulation of **Glucocorticoid** secretion from the adrenal glands.

This rise is Essential for Survival: the absence of glucocorticoids results in **DEATH**.

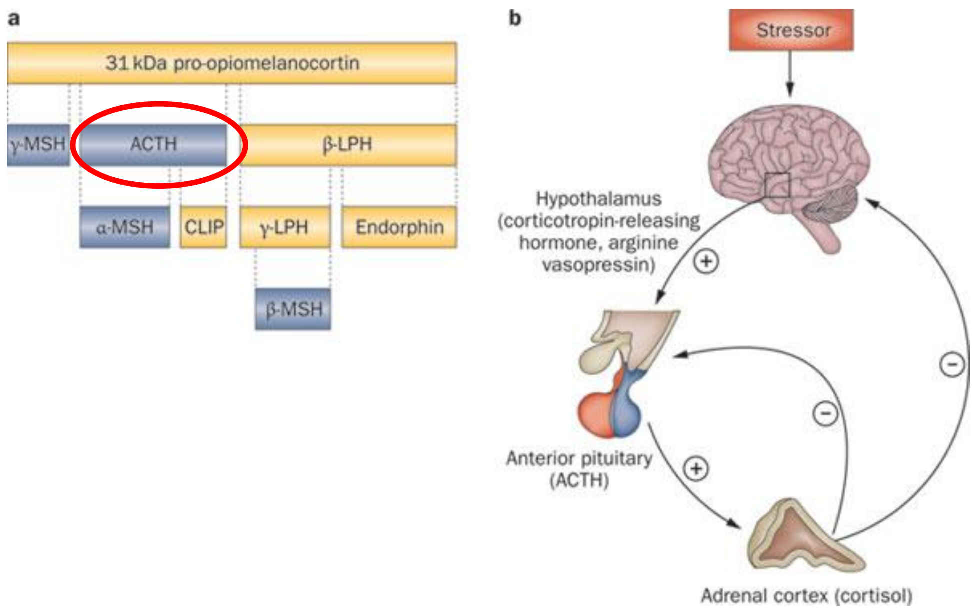
Mechanisms involved in the stress-coping functions of glucocorticoids remain unclear.



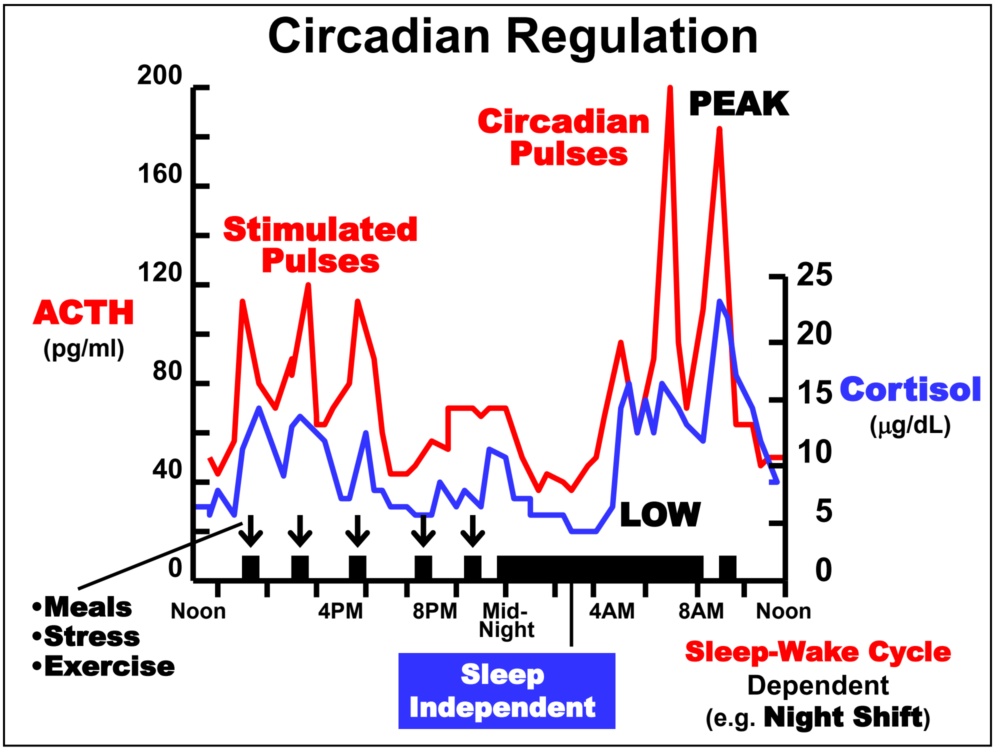
Cortisol secretion occurs through a typical Hypothalamic-Anterior Pituitary axis negative feedback mechanism as diagrammed above. Maintenance of glucocorticoid plasma levels is achieved by the Negative Feedback effects of **Cortisol**, on **CRH** and **ACTH** release at the level of the Hypothalamus and the Anterior Pituitary, respectively (*right arrows*).

Stimulating Factors (*top left*). Basal cortisol secretion level is maintained through constitutive secretion of CRH and ACTH. The balance between these factors fluctuates over the course of a day in response to an intrinsic Circadian Rhythm mediated through higher brain centers. In addition, Stimulated cortisol secretion above basal levels occurs in response to physical or metabolic activities (Exercise, Meals) or non-specific and emotional Stress. Stress induction is apparently mediated, in part, through induction of sympathetic (Epinephrine) stimulation.

ACTH is Biosynthesized as Multi-Peptide Precursor (POMC)



An important feature of **ACTH** is that it is initially synthesized as larger precursor molecule called **Proopiomelanocortin** (**POMC**). POMC is proteolytically cleaved to several active peptides, including ACTH (see Anterior Pituitary Hormone section)



The daily secretion pattern of **ACTH** (*left y-axis*) and **Cortisol** (*right y-axis*) are plotted above in a person with a normal diurnal (sleep-wake) pattern. Note that **Low** levels occur in the late evening (≈ **8 PM – 3 AM**). **Peak** levels in early morning (≈ **5-8 AM**).

Basal levels of both and probably **CRH** (*not shown*) vary according to an intrinsic **Circadian Rhythm**. Circadian spikes and troughs are likely due to fluctuations in the sensitivity of pituitary corticotrophs to CRH during the day (e.g. more sensitive during circadian pulse periods). Physiological purpose remains unclear, but may serve to mobilize energy substrates in anticipation of awakening and increased activity.

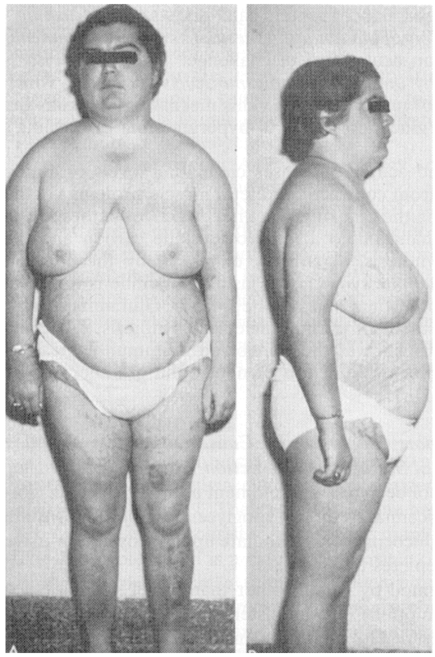
In contrast to growth hormone & prolactin, the appearance of circadian pulses apparently occur **Independent of Sleep** *per se*. That is, they appear whether one is awake or asleep. However, the timing of the circadian pulse spikes can be **<u>shifted</u>** by altering the **Sleep-Wake Cycle** (e.g. working the **Night Shift**).

Irregular **Stimulated Pulses** are superimposed on top of basal levels and circadian bursts. Stimulated pulsatile secretion spikes are initiated by events (*bottom arrows*) such as **Meals**, **Stress** & **Exercise** that are related to the metabolic or other physiological actions of glucocorticoids.

Cushing’s Syndrome/Disease

Cushing’s *Syndrome*:

- **Multiple Etiologies**
- **Anterior Pituitary Adenoma:**
  “Cushing’s *Disease*”
- **2° Hyperglucocorticoidism**
  - **High [ACTH]**
  - **High [CORTISOL]**
- **Adrenal Tumor:**
- **1° Hyperglucocorticoidism**
  - **Low [ACTH]**
  - **High [CORTISOL]**
- **Excess Exogenous GCs:**
- **2° Hyperglucocorticoidism**
  - **High [Glucocorticoid]**
  - **Low [ACTH]**
  - **Low [CORTISOL] (Endogenous)**



Cushing's *Syndrome* is a constellation of classic clinical symptoms and physical features that occur as a consequence of chronic glucocorticoid excess (**Hyperglucocorticoidism)**, which results from Many Causes. The most common form of Cushing's Syndrome (70%) is termed **Cushing's *Disease***. This condition is <u>specifically</u> caused by an unregulatable **Anterior Pituitary Tumor** that secretes excessive **ACTH**, which stimulates **Cortisol Hypersecretion**. Negative Feedback Inhibition of ACTH secreted by the tumor by physiologic levels of cortisol is ineffective toward normalizing secretion.

Clinical laboratory assay of hormone status is diagnostic of a classical **Secondary Hyperglucocorticoid** condition: **<u>High</u> ACTH & <u>High</u> Cortisol**.

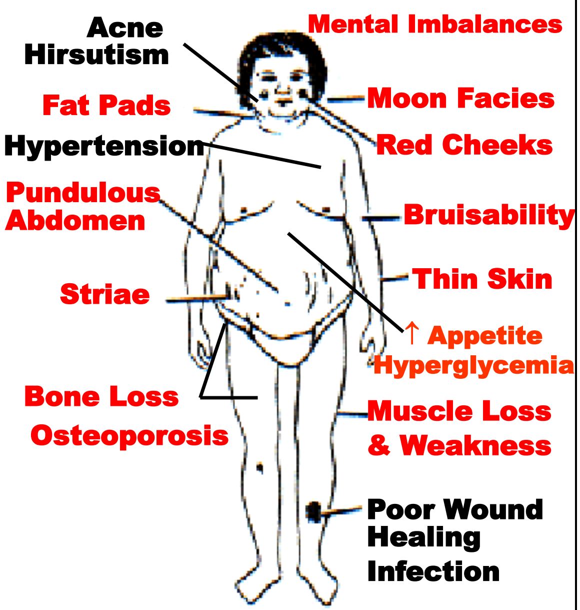
A **Primary Hyperglucocorticoid** resulting in typical Cushing’s Syndrome features is a glucocorticoid-secreting **Adrenal Tumor**. Another **Secondary Hyperglucocorticoid** condition is caused by **Excess Exogenous Glucocorticoid** use. Typical lab findings are shown.

Cushing's Syndrome

Metabolic Effects:
- **↑ Protein Catabolism:**
  - Muscle Loss & Weakness
  - Thin Skin
    - Red Cheeks; Striae
    - Bruisability
- **↑ Bone Catabolism**
  - Bone Loss/Osteoporosis
- **↑ Lipolysis**
  - **–Insulin Insensitivity**:
  - ↑ Appetite (↑ GC = ↓ CRH)
    - Fat Pads; Moon Face
  - Hyper-Glycemia/Insulinism
    - Adrenal Diabetes

Excess Cortisol
- **↑ Mineralcorticoid Activity**
  - Hypertension
- **↑ Androgens**
  - Acne & Hirsutism

Anti-Inflammatory
- Poor Wound Healing
- ↑ Infection; ↓ Immunity



Metabolic Abnormalities: Enhanced glucocorticoid **Protein Catabolism** accounts for **Muscle Loss** and resulting **Muscle Weakness**. Also causes abnormally **Thin Skin** that manifests clinically as facial Plethora (**Red Cheeks**) and abdominal **Striae** in association with Obesity (*see below*). Easy **Bruising** of extremities following minimal trauma is also commonly observed as a consequence of these developments. Similarly, **Bone Loss** caused by excessive **Bone Catabolism** may often result chronically in **Osteoporosis**. Patients commonly present with osteopenia, back pain and sometimes unexplained fractures. Catabolic effects are generally more pronounced in proximal and lower extremities.

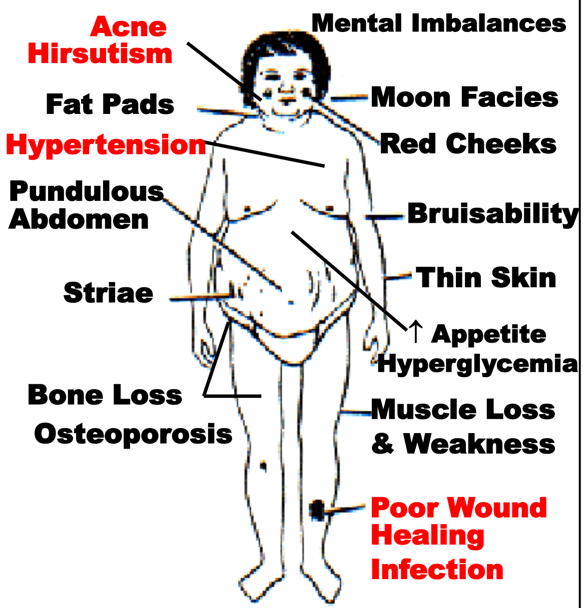
↑ **Lipolysis** & **Insulin Insensitivity** resulting from hyperglucorticoidism causes several associated clinical developments. Actions excessively increase mobilization of fatty acids from adipose tissues with preferential loss of fat deposits from upper and lower extremities. Paradoxically, a common manifestation is Weight Gain with Localized Fat deposition (**Local Obesity**) typically around the face (**Moon Facies**), neck (**Fat Pads**), trunk and abdomen (**Pendulous Abdomen**) with sparing of extremities. Seemingly contradictory to lipolytic glucocorticoid actions, fat accumulation is apparently caused by the ↑ **Appetite** stimulated by the **low CRH** level (CRH normally suppresses appetite) due to the chronic hyperglucocorticoid state (i.e. feedback inhibition) although **Hyperinsulinism** may contribute via its lipogenic actions. Thus, **Hyperglycemia** resulting from overeating and increased hepatic **gluconeogenesis** causes **Hyperinsulinemia** and eventually **Insulin Insensitivity**. Localized pattern is idiopathic. Chronically, a diabetic condition termed **"Adrenal Diabetes"** may develop. Mechanism is similar to that described for Growth Hormone related "Pituitary" Diabetes.

Cushing's Syndrome

Metabolic Effects:
- **↑ Protein Catabolism**
  - **Muscle Loss & Weakness**
  - **Thin Skin**
    - **Red Cheeks; Striae**
    - **Bruisability**
- **↑ Bone Catabolism**
  - **Bone Loss/Osteoporosis**
- **↑ Lipolysis**
  - –**Insulin Insensitivity**
  - **↑ Appetite (↑ GC = ↓ CRH)**
    - **Fat Pads; Moon Face**
  - **Hyper-Glycemia/Insulinism**
    - **Adrenal Diabetes**

Excess Cortisol
- **↑ Mineralcorticoid Activity**
  - **Hypertension**
- **↑ Androgens**
  - **Acne & Hirsutism**

Anti-Inflammatory
  - **Poor Wound Healing**
  - **↑ Infection; ↓ Immunity**



Excess Cortisol Effects: **Hypertension**. Hypertension is a classic feature of Cushing's Disease. Primarily related to intrinsic **Mineralcorticoid** actions of Cortisol and other glucocorticoids. Conditions resulting in excess ACTH secretion may also cause stimulation of excess mineralcorticoid production. Also related to **Vasoconstriction** due to exaggerated **Anti-Inflammatory** actions and tissue hypersensitivity to Catecholamines. **Acne** and **Hirsutism** are the result of **Hyperandrogenism** caused by "shunting" of excess glucocorticoids into the androgen pathway. **Virilism** in females is rare but may be manifested in cases of adrenal carcinoma. However, **Amenorrhea** accompanied by **Infertility** occurs in 75% of premenopausal female patients. Decreased **Libido** is frequent in males and some have decreased body hair and testicular function abnormalities. Related to increased peripheral conversion of excess androgens into Estrogen.

Anti-Inflammatory Abnormalities: Weakened Inflammatory Response due to enhanced anti-inflammatory actions results in **Poor Wound Healing** following even mild **Trauma**. This increases frequency of **Infection**. **Lower Immunity** due to suppression of immunity render patients more susceptible to bacterial and viral diseases. **Mental Imbalances**. Probably related to metabolic and catabolic actions. Mild symptoms include emotional lability and increased irritability. Anxiety, depression, poor concentration and poor memory may also be present. Euphoria is common with occasional manic behavior in some patients. Sleep disorders are also developed as insomnia or early morning awakening. Severe psychological disorders are rare but include severe depression, psychosis with delusions or hallucinations and paranoia. Suicides have been recorded.

Some clinical features of **Cushing's Syndrome (Disease).**

Hypoglucocorticoidism: Addison's Disease

- **Adrenal Destruction/Atrophy**
- **1° Hypoadrenocorticoidism**
  - **↓ [CORTICAL HORMONE]**
    - **Glucocorticoids**
    - **Mineralcorticoids**

    (Androgens Normal)
  - **↑ [ACTH]**

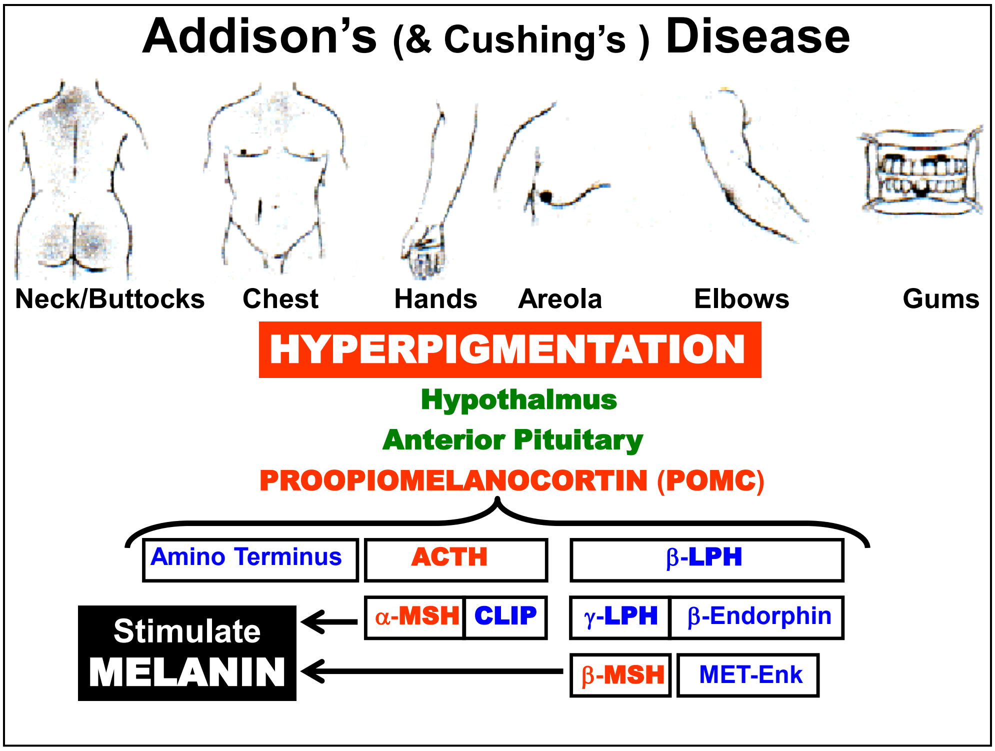
Glucocorticoid Deficiency:
- **Hypotension**
- **Hypoglycemia**
- **Weakness & Fatigue**
- **GI Dysfunction**
- **Appetite & Weight Loss**
- **Nausea & Vomiting**

Mineralcorticoid Deficiency:
- **Hypovolemia**
- **Hypotension**
- **Hyponatremia**
- **Hyperkalemia**
- **Metabolic Acidosis**

The classical example of **Hypoadrenocorticoidism** is <u>total</u> adrenal insufficiency or **Addison's Disease**. Addison's disease is rare (40-60 /1 million) but more common in females than men (2.6:1). However, incidence is predicted to rise as malignant disease patients survive longer and immunodeficiency diseases such as AIDS increases.

Addison's disease most often develops as a result of Adrenal Gland Destruction (e.g., tuberculosis; cancer) or Atrophy (autoimmune disease). In such cases, Addison's is classified as a **Primary (1°) Hypoglucocorticoidism** condition. Laboratory diagnosis is based on **(↓) [Cortical Hormone]** (Glucocorticoids & Mineralcorticoids) and **(↑) ACTH** levels. Androgen deficiency is rare because of Gonadal Steroids compensation.

Clinical features related to Glucocorticoid Deficiency (*top right*) are generally <u>Opposite</u> Cushing's Disease, but are more subtly diagnostic of the disease. Mineralcorticoid Deficiency (*bottom right*) causes classical manifestations related to the normal actions of the hormone in plasma volume and electrolyte homeostasis: (see Aldosterone section). Patients require life-long administration of Mineralcorticoids & Glucocorticoids.



Hyperpigmentation is a common clinical feature of Addison's Disease. It manifests due to chronically increased secretion of **ACTH** which is synthesized as a common peptide (Proopiomelanocortin **(POMC)** see Pituitary Gland) that encodes several active peptides, including Melanocyte Stimulating Hormone (**MSH**).

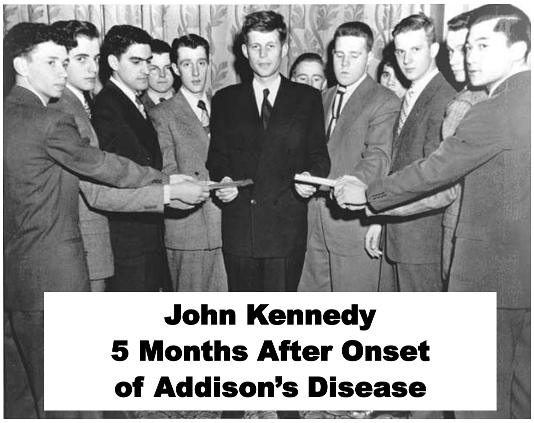
MSH stimulates Pigment Secretion from melanocytes in the skin (predominantly animals). High levels of ACTH and the relatedness of the peptides results in direct cross stimulation of pigmentation by ACTH and/or increased processing of excess ACTH to MSH.

As illustrated above (*top*), Melanin is not always distributed evenly, occurring in blotches and especially in the thin skin areas and mucus membranes of the Hands, Elbows, Gums & Areola of Nipples.

However, hyperpigmentation is **<u>NOT Specific to Addison's disease</u>**. For example, since hyperpigmentation is the result of hypersecretion of ACTH, the symptom c an develop in association with certain etiologies of Cushing's Syndrome (e.g. Cushing's Disease: 2° Hyperglucocorticoidism) caused by an anterior pituitary adenoma. Moreover, hyperpigmentation is may also be associated with other diseases.

Hyperpigmentation feature related to glucocorticoid disorders.

Thomas Addison



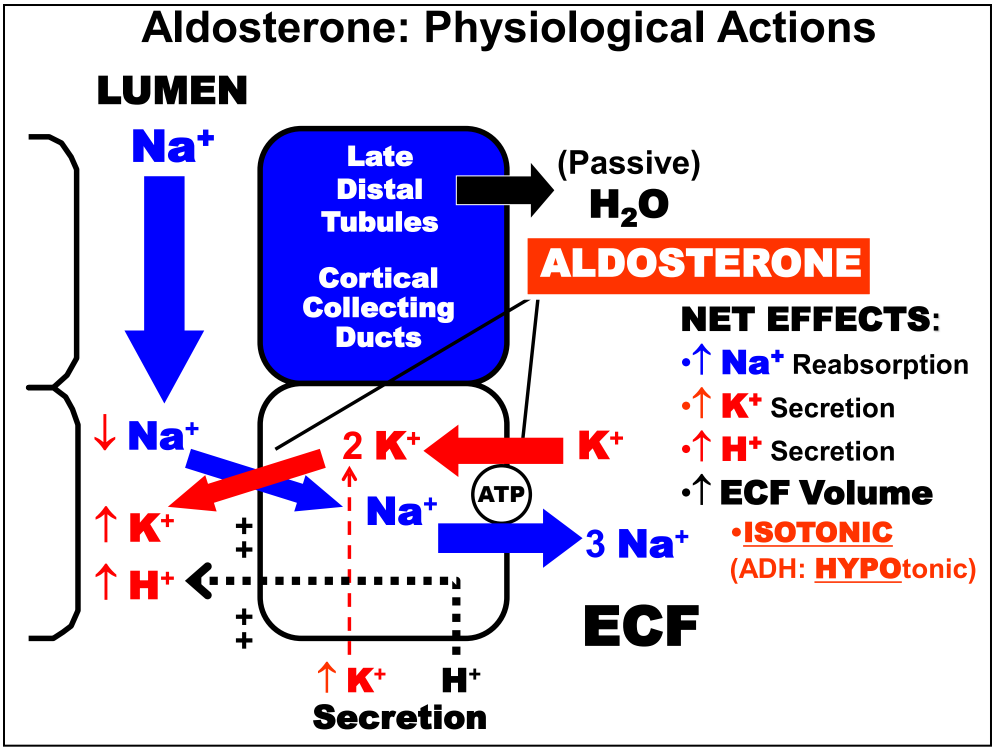
John Kennedy
5 Months After Onset
of Addison's Disease

John Kennedy had Addison's Disease

Disease named for physician Thomas Addison who characterized the disease.

President John F. Kennedy developed Addison's Disease.

Mineralcorticoids

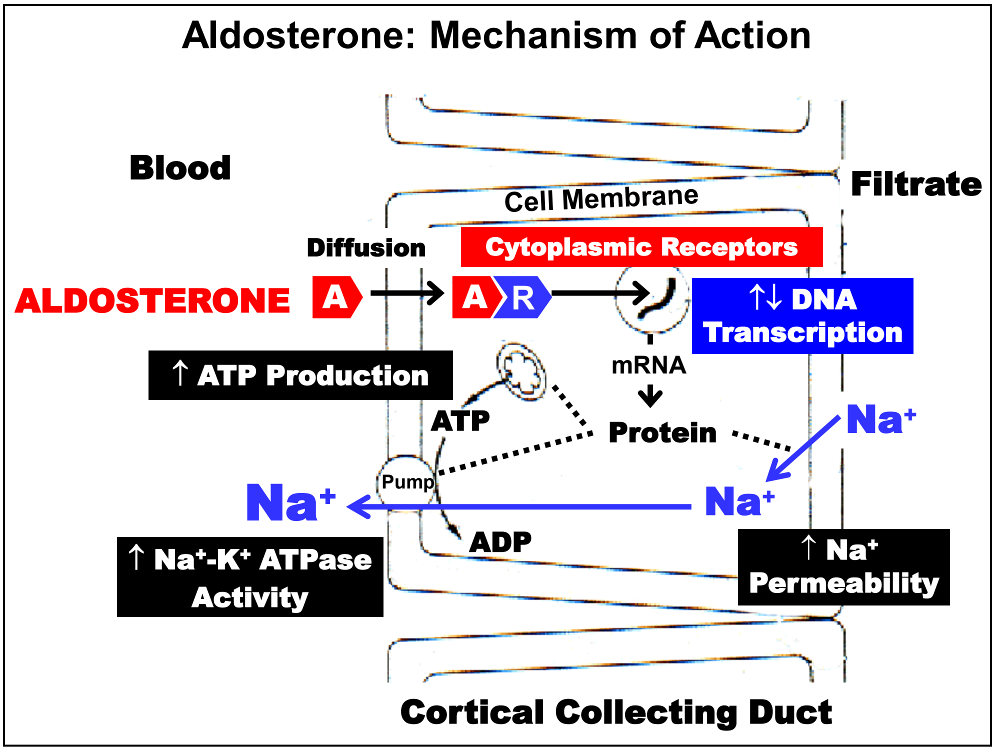


Although several adrenal cortical hormones possess **Mineralcorticoid** activity, **Aldosterone** is by far the most physiologically important.

The physiological functions of aldosterone were discussed during the Renal System unit of IS_CVRR. A brief summary is provided above for review.

Aldosterone acts primarily on the **Late Distal Tubules & Cortical Collecting Ducts** and has the Net Effects shown to the right.

Note that in contrast to ADH, aldosterone does NOT directly influence plasma osmolality because of its reabsorption of an **Isotonic** fluid volume. This is an important contrast to the action of **ADH**, which results in **HYPOtonic** fluid reabsorption.

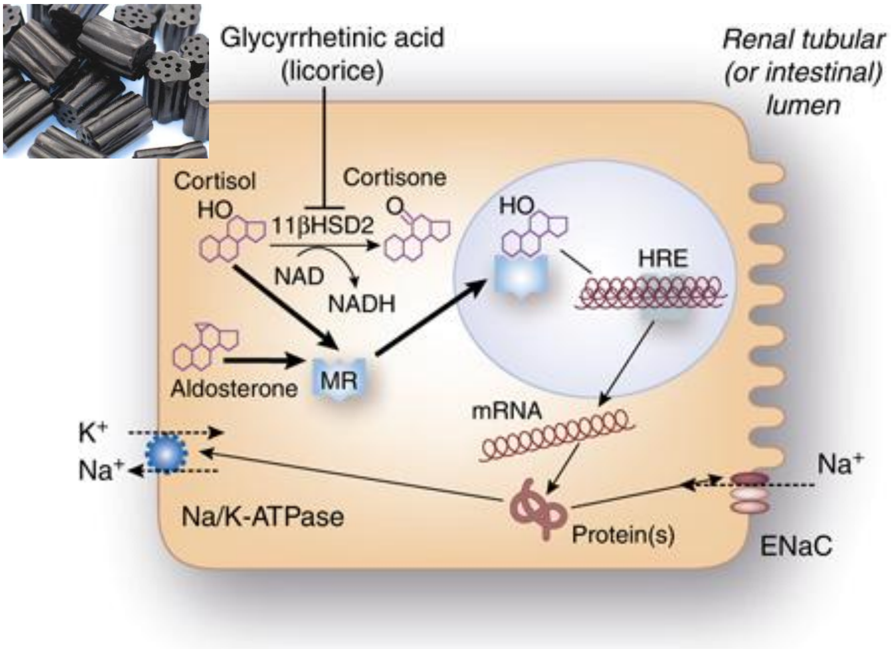


The mechanism of action of **Aldosterone** on target cells is very similar to that described for Glucocorticoids. (*top left to right*)

Lipid soluble Aldosterone (**A**) enters target cell (Cortical Collecting Duct) by **Diffusion** and exerts its actions by binding to Intracellular **Cytoplasmic Receptors** (**R**).

The resulting aldosterone-receptor complex (**A-R**) migrates to the nucleus where mainly affects (↑↓) **DNA Transcription**. Synthesized **Proteins** stimulate renal **Na+ Reabsorption** by affecting cellular various processes indicated above.

Mineralcorticoid Receptor Competition by Cortisol

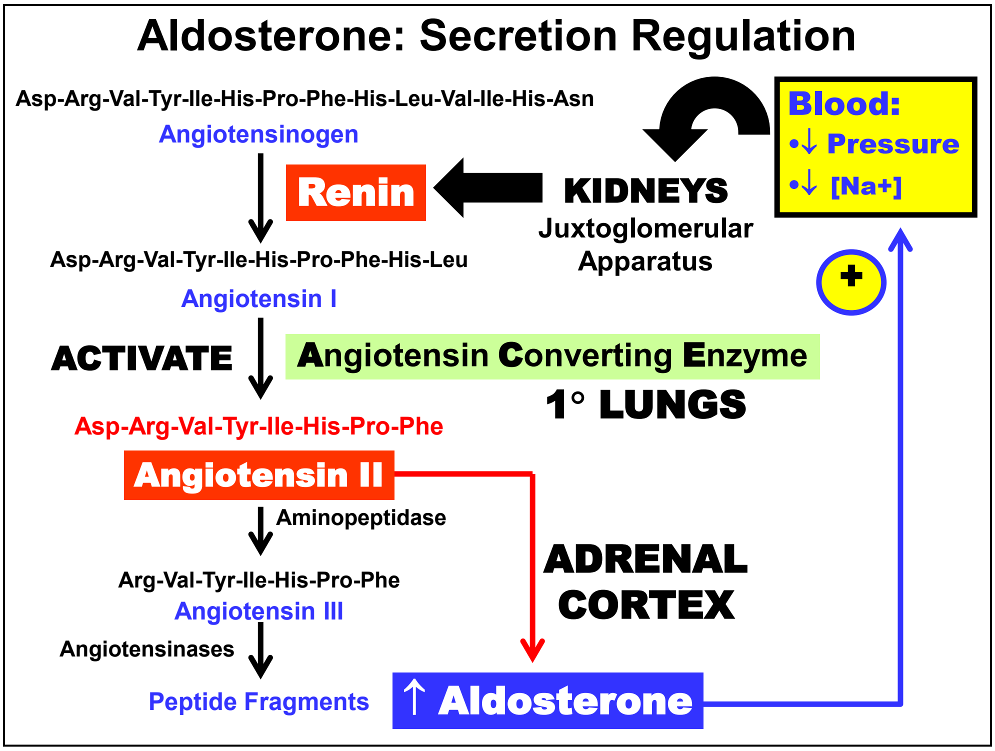


Physiologically, **<u>mineralcorticoid receptors (MR) bind Cortisol with nearly equal affinity</u>**.

However, even though cortisol plasma concentration (14 mg/dL) is much greater than aldosterone (0.006 mg/dL), only aldosterone's actions are normally present. Mineralcorticoid specificity is conferred by the enzyme **11β-Hydroxysteroid Dehydrogenase-2** (**11β-HSD2**), which converts Cortisol to **Cortisone** (not recognized by the receptor).

Aldosterone is unaffected by 11β-HSD2, which is present at high concentrations in renal tubules. Persons with a genetic deficiency of 11β-HSD2 suffer from symptoms of mineralcorticoid excess (hypertension & hypokalemia) as a result of overstimulation by cortisol (Pseudohyperaldosteronism).

Interestingly, an acquired impairment may also develop after ingestion of excessive amounts of **Licorice** (real, not Twizzler®) which contains an <u>inhibitor</u> of 11β-HSD2 : **Glycyrrhetinic Acid**



The **Renin-Angiotensin II System** is the Major Regulator of Aldosterone secretion. The Renin-Angiotensin II pathway regulates Aldosterone secretion from the Adrenal Cortex by a negative feedback inhibition mechanism. These factors, in turn, are ultimately regulated by changes in Blood Pressure & NaCl Flux across the Juxtaglomerular Apparatus of the **Kidneys**.

Renin is a very specific protease secreted from the Juxtaglomerular Apparatus of the kidneys that catalyzes the formation of the decapeptide **Angiotensin I** from the precursor **Angiotensinogen** secreted by the liver. Both Angiotensinogen & Angiotensin I are biologically <u>Inactive</u>. Finally, Angiotensin II is activated by **Angiotensin Converting Enzyme** (**ACE**), which cleaves two amino acids from Angiotensin I as it passes through the pulmonary circulation (Lungs) and across the glomerular capillaries. Finally, Angiotensin II circulates to the **Adrenal Cortex** where it stimulates **Aldosterone** secretion. Aldosterone then acts to raise ECF volume and blood pressure (*right upward arrow*).

Mineralcorticoid Disorders

DEFICIENCY:	EXCESS:
•1° HYPOaldosteronism:	•1° HYPERaldosteronism
•Adrenal Destruction	•Adenoma (Glomerulosa)
•Addison's Disease	•Conn's Syndrome
•Clinically:	•Clinically:
•Hyponatremia	•Hypernatremia
•Hyperkalemia	•Hypokalemia
•Mild Acidemia	•Mild Alkalosis
•Hypotension	•Hypertension
•↓ ECF Volume: **Shock**	•↑ ECF Volume
•Death (Days-Weeks)	•Low Renin
•Treatment:	•Treatment:
•Mineralcorticoids	•Adenoma Removal
•NaCl	•Andrenalectomy

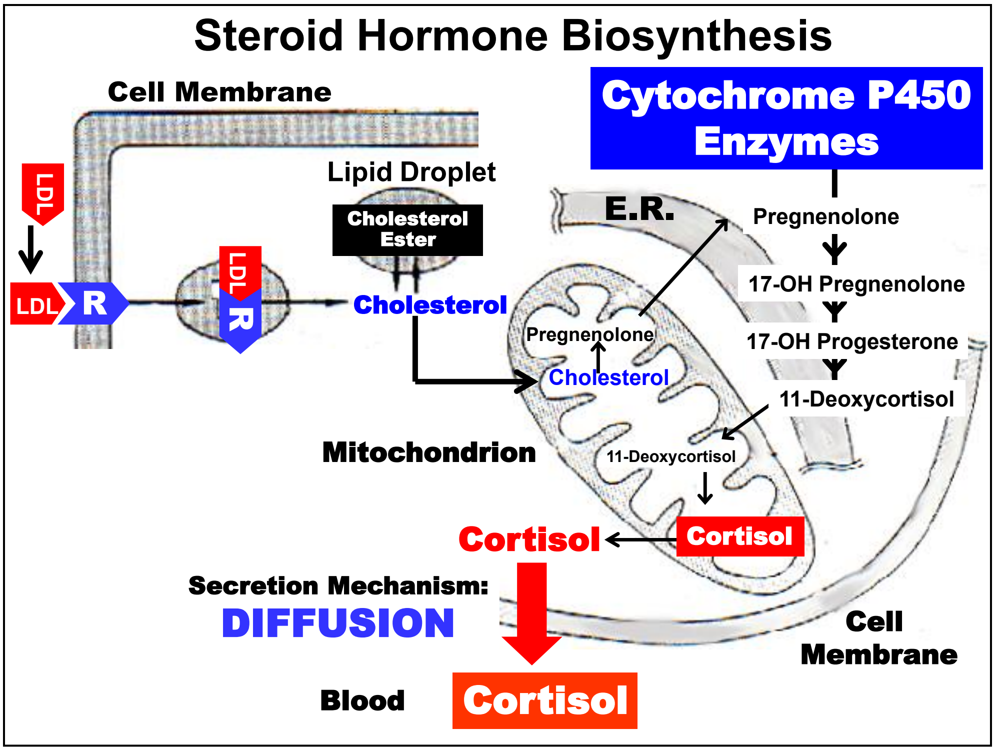
__Mineralcorticoid Deficiency__: __Primary Hypoaldosteronism__ (*left*).

Adrenocortical Insufficiency resulting from destruction or atrophy of the adrenal cortex is the most common cause of mineralcorticoid deficiency (e.g. Addison's disease). Clinically classified as **Primary Hypoaldosteronism** conditions. Major Clinical Manifestations are listed. How are these related to the physiological functions of mineralcorticoids? Lack of aldosterone is **FATAL** if left untreated. Death occurs within days-weeks mainly due to **Hypotension**, which cause **Circulatory Insufficiency** that leads to **Shock**. Treatment involves therapeutic **Mineralcorticoids** and increased dietary or infused **NaCl**.

__Mineralcorticoid Excess__: __Primary Hyperaldosteronism__ (*right*).

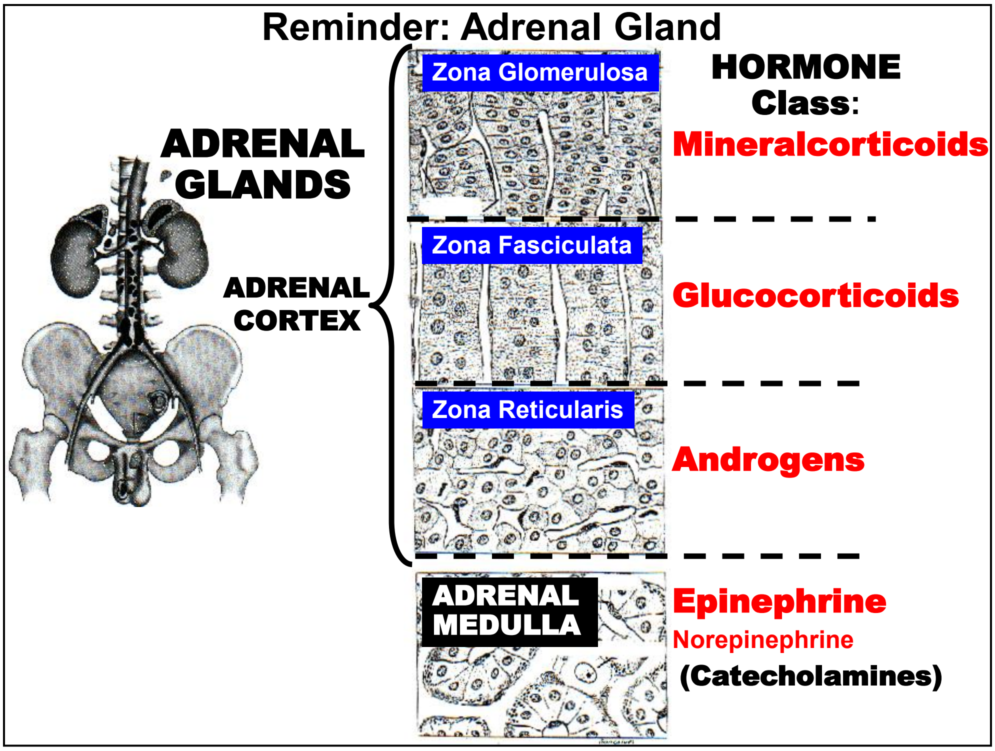
Aldosterone-secreting Glomerulosa Adenomas are the most common cause (75%) of mineralcorticoid excess. Clinically classified as **1° Hyperaldosteronism** condition or ***Conn's Syndrome***. . Major Clinical Manifestations are listed. Decreased plasma Renin level is a classic feature but is not itself diagnostic. Not as immediately life-threatening as hypoaldosteronism, but serious manifestations from **Hypertension** development. Treatment is either Tumor Removal or Adrenalectomy.

Narrated PowerPoint Presentations



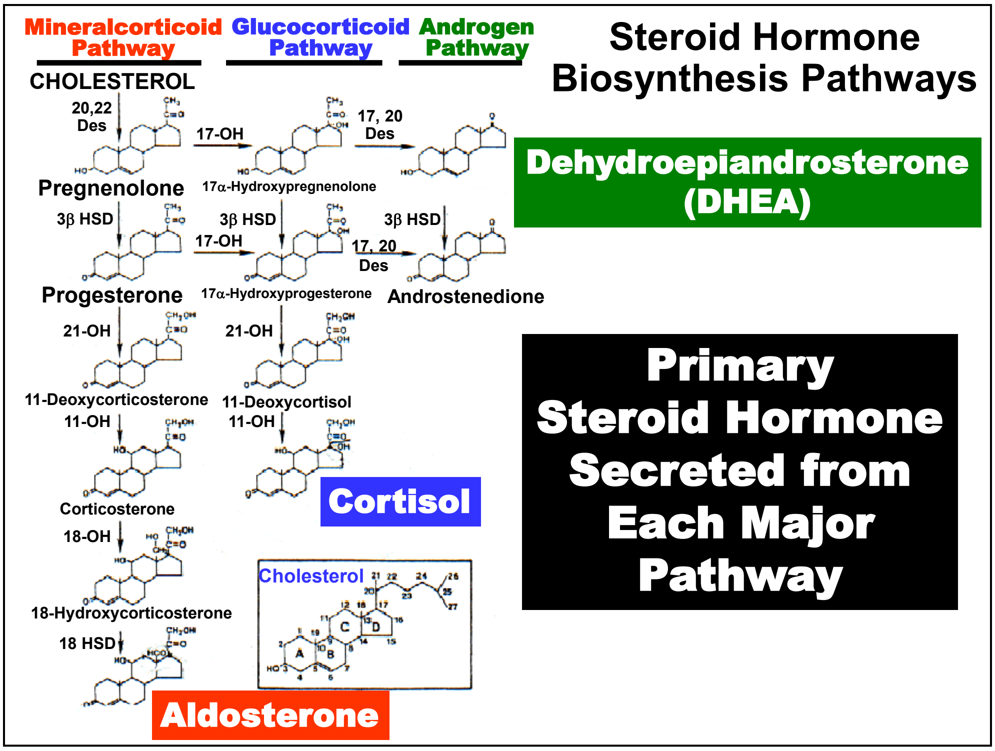
Steroid hormone biosynthesis (e.g. **Cortisol**) begins with receptor-mediated endocytosis (**LDL-R**) of the precursor **Cholesterol** into the cell within low-density lipoprotein (**LDL**) particles of lysosomes (**LDL-R** oval-*top left*). Cholesterol can also be liberated from Cholesterol Esters, its intracellular storage form, within Lipid Droplets. In either case, free Cholesterol is first transported into Mitochondria (*center*) where a 6-carbon fatty acid is enzymatically cleaved to form Pregnenolone, which is then transported to the endoplasmic reticulum (**ER**). (*top right*) Within the ER, Pregnenolone is converted to 11-Deoxycortisol through a series of steps catalyzed by a series of specific **Hydroxylase** enzymes. The intermediate 11-Deoxycortisol is next returned to Mitochondria for final modification to the active hormone **Cortisol**, which is secreted into the Blood by Diffusion (*bottom center*).

Although Cortisol synthesis is shown above, understand that the particular class of steroid (e.g Gluococorticoid) as well as the specific steroid hormone (e.g. Cortisol) produced is dependent upon the complement of **Cytochrome P450** (**P450c**) enzymes available within the adrenal cell. This is summarized next.



Reminder that the adrenal cortex is subdivided into 3 zones that secrete the 3 major classes of cortical hormones: The Zona Glomerulosa (*upper zone*), exclusively produces **Mineralcorticoid** hormones. The Zona Fasciculata (*middle zone*) primarily produces **Glucocorticoids** and the Zona Reticularis (*bottom zone*) almost exclusively produces **Androgens**.

Specificity of regional secretion of these hormones is determined biochemically by the complement of P450 enzymes predominantly expressed within each region.



Enzymatic synthesis pathways for cortical steroid hormones are shown above. To produce a specific steroid class (i.e. Mineralcorticoids; Glucocorticoids; Androgens), Pregnenolone, the first steroid intermediate formed from Cholesterol is shunted through one of the three major enzyme pathways (*top*).

The predominant pathway used by each zone of the adrenal cortex depends on the availability or predominance of specific sets of **P450 Enzymes**.

Cortisol is the principal **Glucocorticoid**.

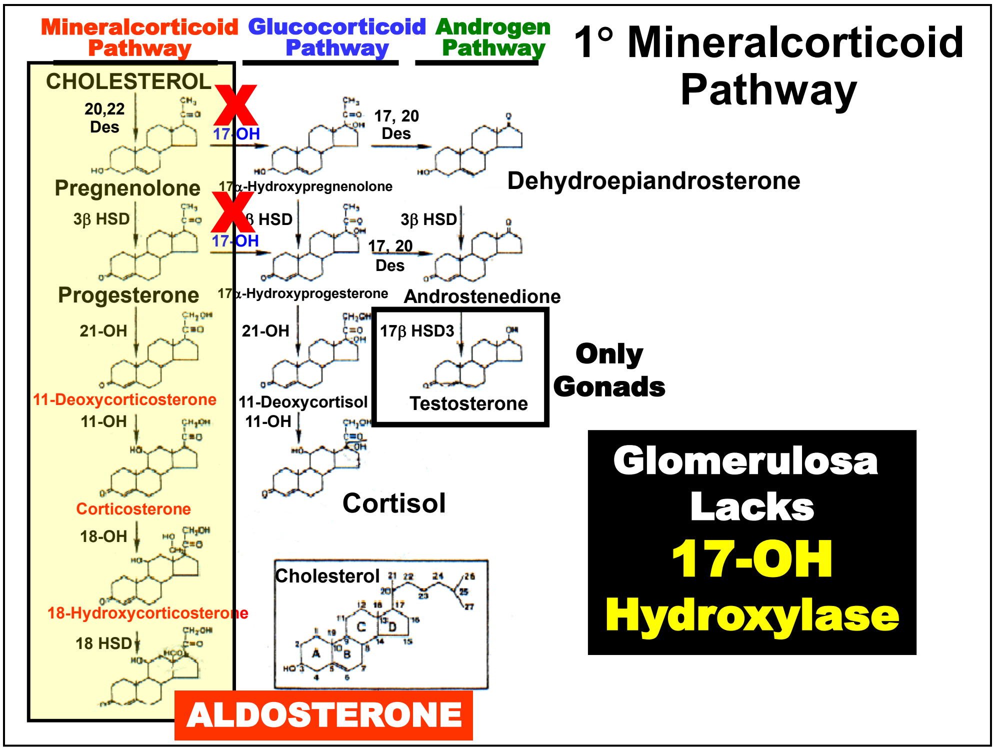
Cortisol is secreted primarily from the **Fascilulata** layer of the Adrenal Cortex.

Aldosterone is the principal **Mineralcorticoid**.

Aldosterone is secreted exclusively from the **Glomerulosa** layer of the Adrenal Cortex.

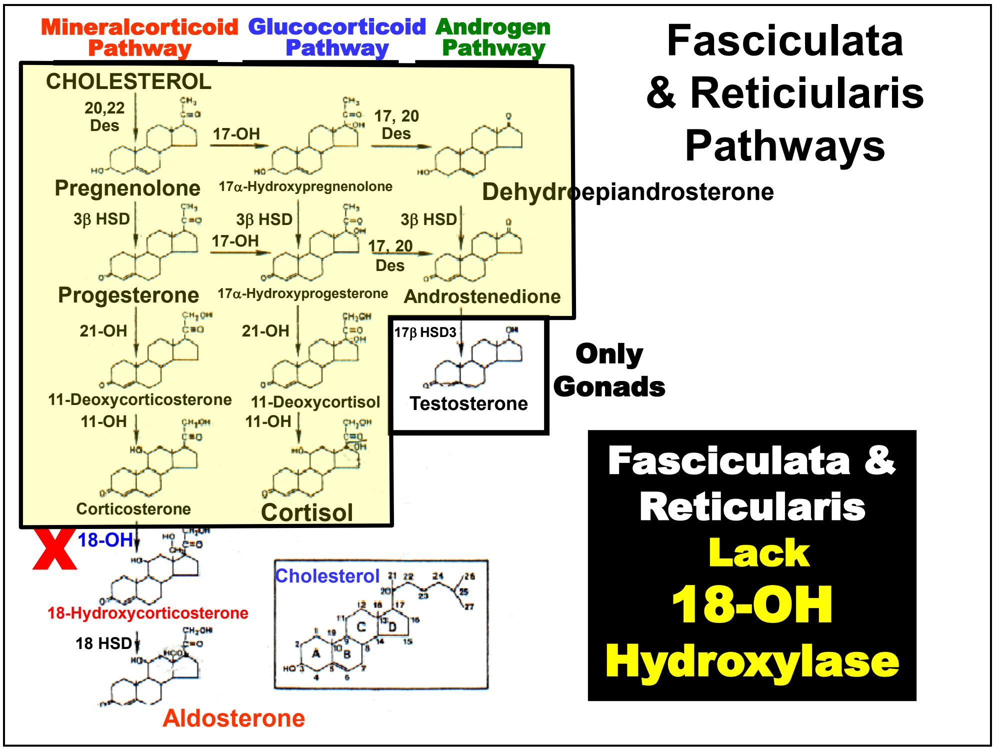
Dehydroepiandrosterone is the prominent adrenal **Androgen**.

DHEA is secreted primarily from the Reticularis layer of the Adrenal Cortex.



The predominant enzymatic pathway utilized within the **Glomerulosa** region is of the adrenal cortex indicated by the shaded area. The major secretion product is the mineralcorticoid hormone **Aldosterone**.

Due to the virtual absence of the **17-OH Hydorxylase** P450 enzyme in this region, the glomerulosa layer is unable to synthesize Glucocorticoid or Androgen hormones to any significant extent and is restricted to the biosynthesis of mineralcorticoid pathway products.

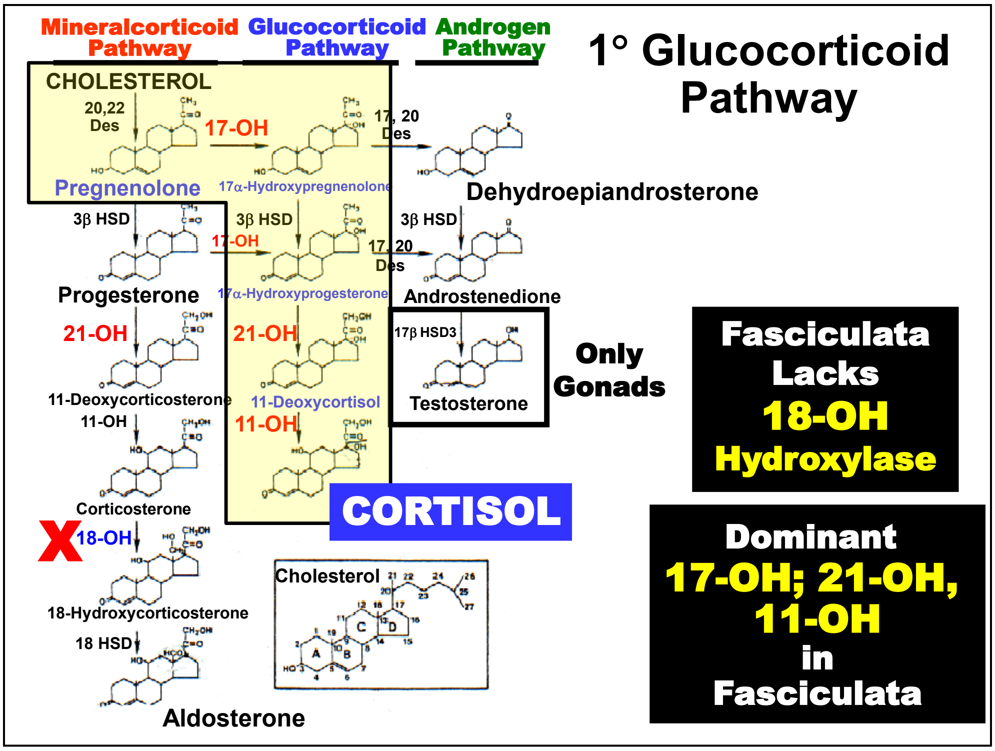


Fasciculata & Reticularis Biosynthesis Pathways.

Both layers lack **18-Hydroxylase (18-OH)** enzyme activity. Thus, neither layer can biosynthesize **Mineralcorticoids** (i.e. **Aldosterone**).

The Fasciculata & Reticularis layers can each <u>POTENTIALLY</u> biosynthesize both **Glucocorticoid & Androgen** steroids (*boxed area*).

However, each layer follows a predominant pathway because of the predominance or lack of specific enzymes. This is summarized next.

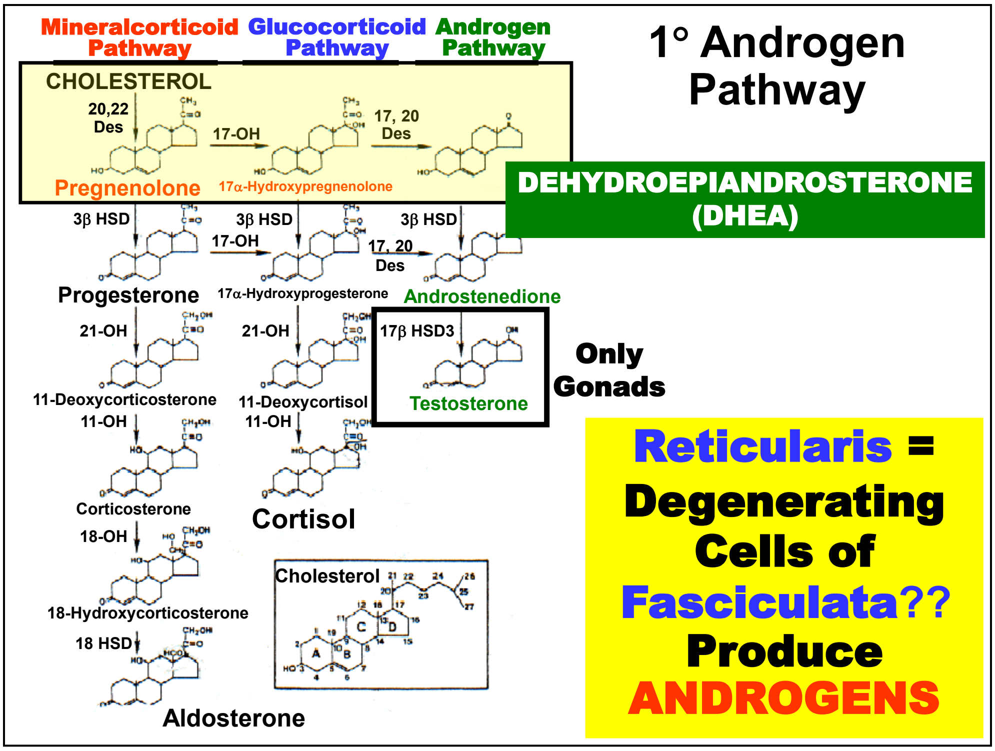


The predominant enzymatic pathway utilized within the **Fasciculata** region is of the adrenal cortex indicated by the shaded area. The major secretion product is the glucocorticoid hormone **Cortisol**.

Due to the absence of the **18-OH Hydroxylase** enzyme in this region, the <u>fasciculata layer is normally unable to synthesize Mineralcorticoid hormones</u>.

Although fasciculata cells are capable of producing Androgens (e.g. DHEA), they are not significantly produced by this region physiologically due to the dominant **17-OH & 21-OH,** and **11-OH Hydroxylase** enzymes in this region.

However, certain clinical diseases or genetic disorders that affect relative adrenal cortex enzyme activity may result in excessive production of adrenal androgens, which is used diagnostically to identify such conditions.



The predominant enzymatic pathway utilized within the **Reticularis** region is of the adrenal cortex indicated by the shaded area. The major secretion product is the weak androgenic hormone **Dehydroepiandrosterone (DHEA)**.

Similar to the fasciculata layer, the **18-OH Hydroxylase** enzyme is absent from this region and is normally unable to synthesize Mineralcorticoid hormones.

However, unlike the fasciculata layer, the **21-OH Hydroxylase** enzyme activity in the reticularis region is very weak and does not typically synthesize significant amounts of Glucocorticoids.

Thus, the reticularis layer is generally thought to be an inner layer of degenerating fasciculata cells whose default pathway results in predominantly androgen production.

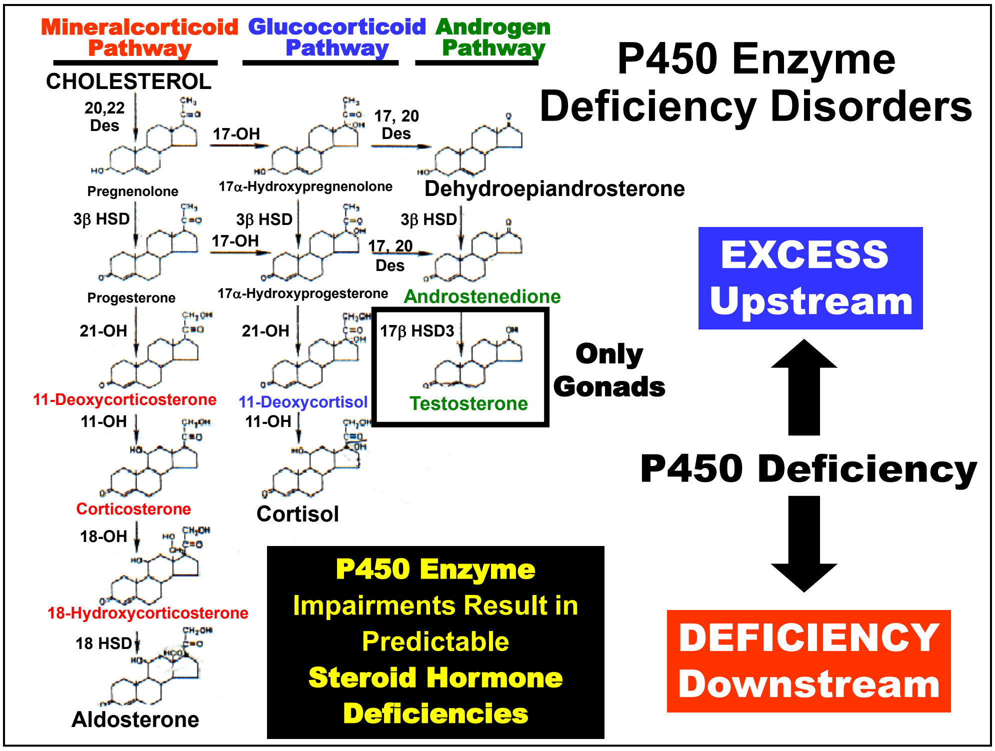
Congenital Adrenal Hyperplasia:

Specific P450 Enzyme Deficiencies

Differential Diagnosis

- **Identify Enzyme Action in Each A.C. Zone**
- **Impact of Enzyme Deficiency on Normal Hormone**
- **Which Abnormal Hormones are Secreted Instead?**
- **Predict Patient Presentation**

Steps in analysis of **Congenital Adrenal Hyperplasia (CAH)** conditions.



A general rule of thumb (*right*) is that with the development of a **P450 Enzyme Deficiency**, steroid intermediates will become **Deficient Downstream** of the deficiency and will accumulate in **Excess Upstream** of the impaired site of action of the enzyme.

This concept is illustrated in the next example involving the congenital absence of the P450 **21-Hydroxylase** enzyme.

NOTE: To assist in rationalizing the following examples, it may be helpful to understand that steps in the steroid pathway are **Non-Reversible**. That is, once an "upstream" intermediate is converted to a "downstream" product, it cannot be re-formed into the upstream steroid product (note ***one-way arrows***). For example, once 11-Deoxycorticosterone (Mineralcorticoid Pathway) is converted to Corticosterone by the 11-Hydroxylase enzyme, Corticosterone cannot be converted back to 11-Deoxycorticosterone. Therefore, if 11-Hydroxylase function becomes impaired within the glomerulosa layer, 11-Deoxycorticosterone level would be predicted to be the predominant hormone secreted from that layer whereas all downstream intermediates (18-Hydroxycorticosterone) and end-products (Aldosterone) would become deficient as they are unable to be synthesized. Similar dynamic for the Glucocorticoid Pathway (i.e. Fasciculata layer), which would secrete 11-Deoxycortisol instead of Cortisol. In both cases, diagnostic of an 11-OH Deficiency.

Overlapping Steroid Class Actions

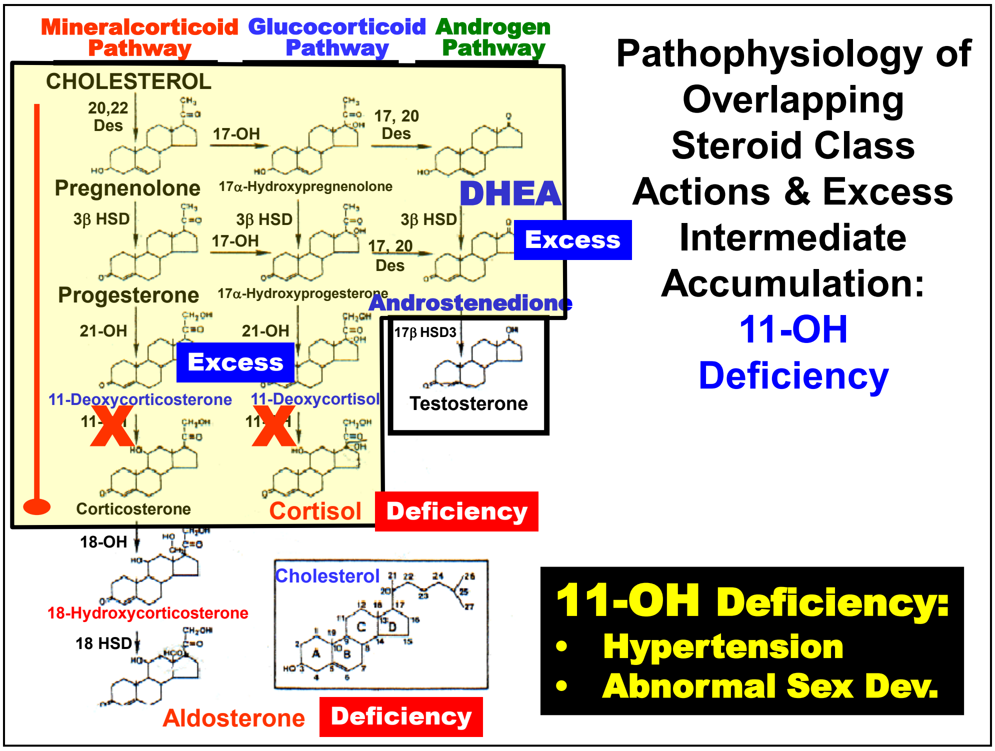
Steroid	Average [Plasma] (μg/dl)	Glucocorticoid Activity	Mineralcorticoid Activity
Cortisol	14	100	0.03
Corticosterone	0.4	30	0.5
Aldosterone	0.006	30	100
11-Deoxycorticosterone	0.006	20	3.3
Dehydroepiandrosterone	17.5	0	0

Clinical Importance of Overlapping Actions:
Aldosterone Deficiency = ↑↑ 11-Deoxycorticosterone

It is important to realize that most **Glucocorticoids** possess some although weak Mineralcorticoid Activity and several **Mineralcorticoids** have at least some and often significant Glucocorticoid Activity. For example, note that while **Aldosterone** has the strongest mineralcorticoid activity (set to **100%**), but also possess about **30%** the relative glucocorticoid activity of the most potent glucocorticoid, **Cortisol** (**100**)%. Conversely, **Cortisol** possesses some, albeit weak, mineralcorticoid activity relative to **Aldosterone**.

CLINICAL NOTE: Overlapping steroid actions come into play clinically when production of one steroid becomes deficient and another excessive. For example, although **Deoxycorticosterone** possesses much weaker mineralcorticoid activity relative to **Aldosterone** (*see above*), it **<u>competes for the same receptor as Aldosterone in the kidneys</u>**. Disorders causing **<u>decreased</u> Aldosterone** production, but **greatly <u>increased</u> Deoxycorticosterone** production can result in competitive blockage of aldosterone action at its own receptor. Since unlike Aldosterone, <u>11-Deoxycorticosterone is NOT regulated by the renin-angiotensin negative feedback mechanism</u>, the result is excessive renal Na+ reabsorption. That is, the receptor is under constitutive stimulation by deoxycorticosterone_. Often, excessive Na+ retention can lead to **Hypertension & Hypokalemia** despite development of **(Secondary) Hypoaldosteronism**.

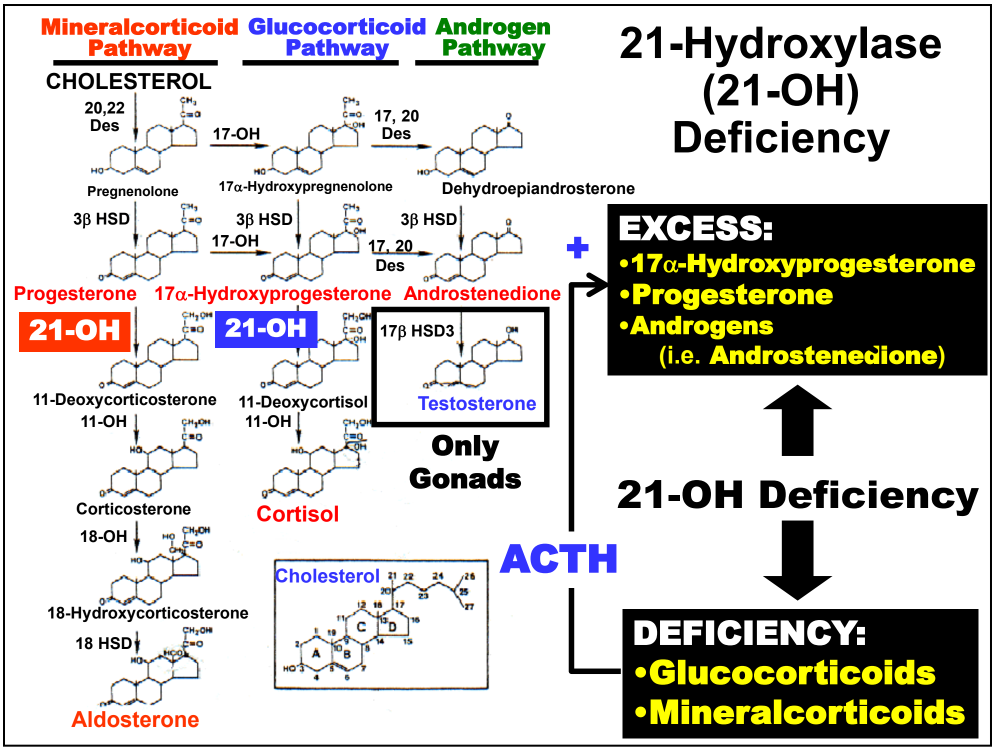
i.e. **<u>patients display the clinical symptoms of Hyperaldosteronism, but pardoxically possess Low Aldosterone levels</u>**.



e.g. Loss of 11-OH activity results in accumulation of intermediates (11-Deoxycorticosterone and 11-Deoxycortisol), which are normally secreted at low levels physiologically.

Pathophysiologically, two consequences of the deficiency result in the observed clinical manifestations: (1) accumulation of products just upstream from the site of enzyme action may have overlapping steroid class effects as described earlier. In this case, accumulation of the weak mineralcorticoid, **11-deoxycorticosterone**, commonly causes development of **Hypertension**; and (2) accumulation of steroid substrate upstream from the deficiency results in excess production of normal steroid products. In this case, the inability to metabolize 11-deoxycortisol and 11-deoxycorticosterone to cortisol and aldosterone, respectively, results in shunting of upstream intermediates (11a-Hydroxyprogesterone and 17a-Hydroxypregnenolone) into the androgen pathway resulting in **excessive DHEA** and **Androstenedione** synthesis. Clinically, this may cause **Abnormal Sexual Development** of fetuses.

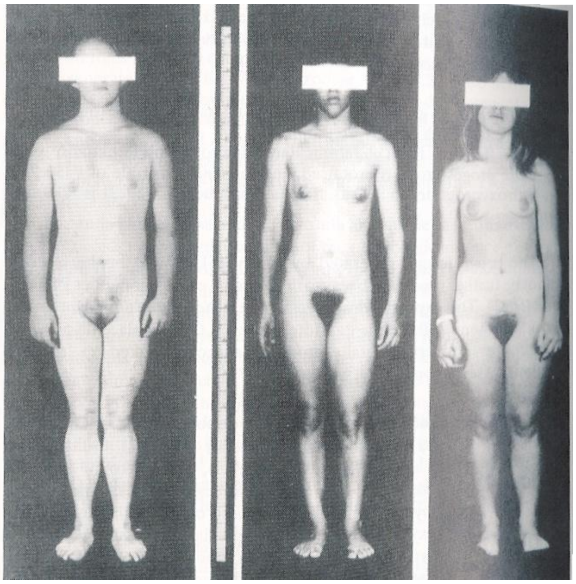
This concept is used diagnostically to distinguish certain P-450 enzyme deficiency disorders from others.



The sites of action of the 21-Hydroxylase enzyme in the steroid biosynthesis pathway are highlighted above (*bolded*). As indicated, the absence of 21-Hydroxylase action would result in impaired Glucocorticoid and Mineralcorticoid synthesis (Downstream Deficiency). Moreover, because of the inability to convert intermediates immediately upstream of 21-Hydroxylase's actions to downstream products such as 11-Deoxycortisol, those steroids (17 α-Hydroxyprogesterone, Progesterone & Androgens) accumulate (Upstream Excess).

This effect is amplified tremendously because the Hypothalamic-Pituitary Feedback mechanism that regulates Cortisol production quickly senses the deficient Cortisol levels and attempts to simulate more production through **ACTH** secretion (*center*). A vicious cycle begins that results in compensatory hypertrophy of the adrenal cortex (adrenal hyperplasia) but continued deficiencies of Mineralcorticoid & Glucocorticoid hormones and the accumulation of male Androgens (primarily Androstenedione). Major clinical consequences are summarized in the next figure.

Congenital Adrenal Hyperplasia: 21-OH Deficiency



Clinically:

Females:

- **Masculinization**

Males:

- **Precocious Puberty**

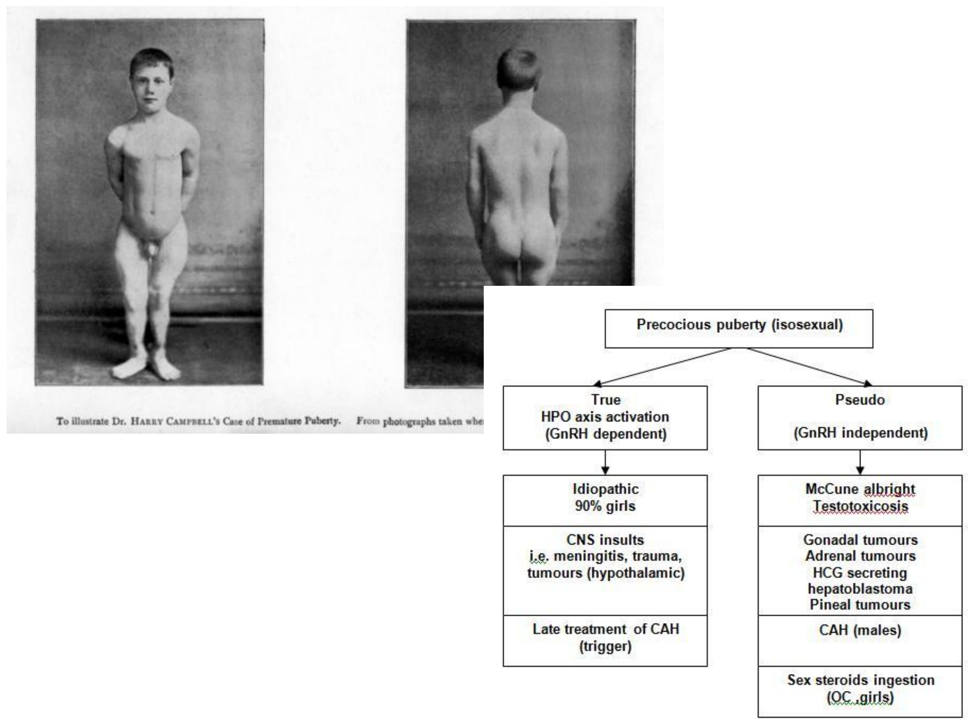
Onset, Salt-Wasting & Hypoglucocorticoidism VARIABLE (≈Severity of Mutation)

Treatment: Exogenous Steroid Replacement

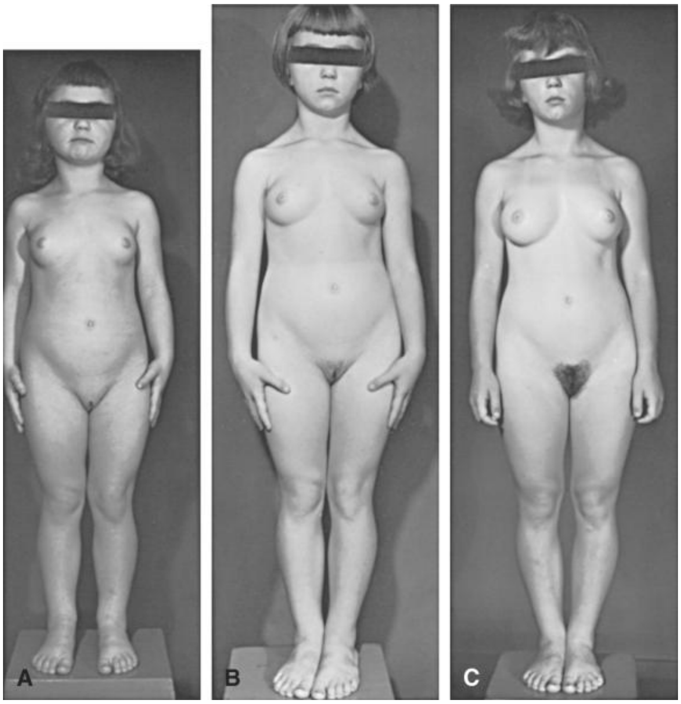
(*top right text box*) The most severe forms of this deficiency are usually fatal *in utero* or shortly after birth if left untreated. However, patients often reach maturity with milder forms. In these cases, the most consistent and outstanding clinical manifestation in **Females** is **Masculinization** of primary and secondary sexual characteristics. In **Males**, **Precocious Puberty** occurs. Both are related to excess **Androgen** accumulation.

(*middle right text box*) Expected clinical manifestations related to Mineralcorticoid & Glucocorticoid deficiencies are actually quite **Variable** and are related to the Severity of the Mutation affecting the enzyme. **Salt-Wasting** (Aldosterone), poor wound healing & stress-coping (Cortisol) are the major predictable clinical developments. Some of these will become more apparent after discussion of the physiological role of these hormones.

(*bottom right text box*) Treatment (Exogenous Steroid Replacement) is aimed at compensation for deficient hormones. However, an equally important effect is that steroid replacement terminates the ACTH stimulatory cycle that causes Hyperandrogenism.



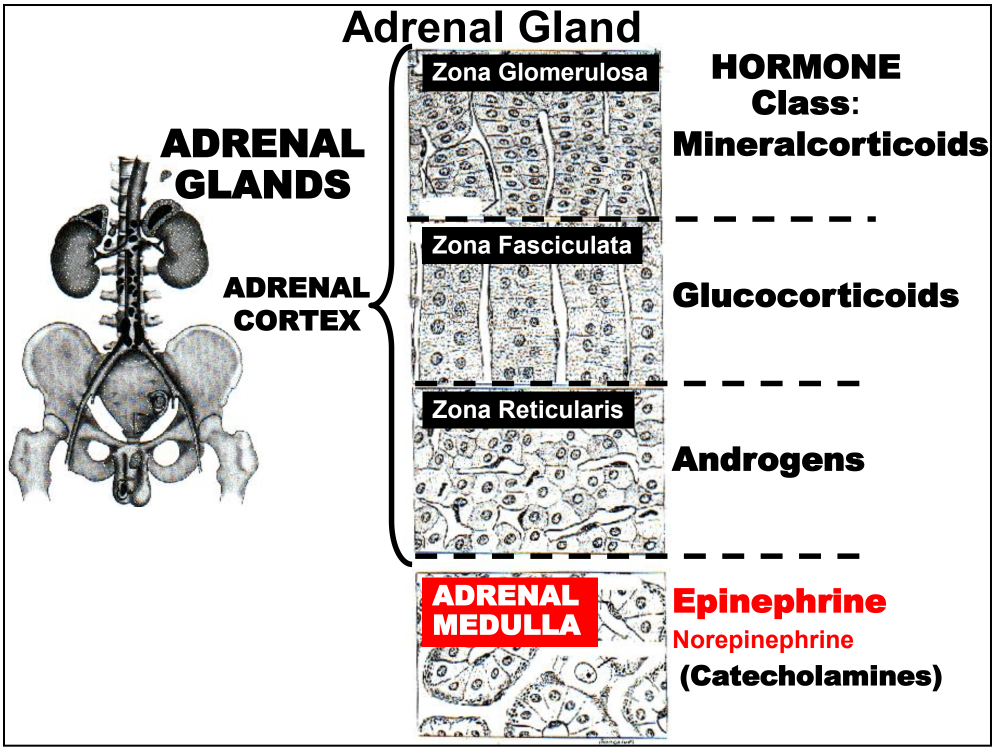
Congenital Adrenal Hyperplasia: **21-OH Deficiency** (**Male Precocious Puberty**)



Congenital Adrenal Hyperplasia: **21-OH Deficiency** (**Female Precocious Puberty**)

Narrated PowerPoint Presentations

1. **Overview & Principles of Endocrine System**
2. **Hypothalamic Neuroendocrine Hormones**
3. **Anterior Pituitary Hormones**
4. **Posterior Pituitary Hormones**
5. **Thyroid Hormones Basic Properties**
6. **Thyroid Hormone Disorders**
7. **Parathyroid Hormone**
8. **Vitamin D and Calcitonin**
9. **Adrenal Cortex Hormones**
10. **Steroid Hormone Biosynthesis & CAH Disorders**
11. **Adrenal Medulla Hormones**



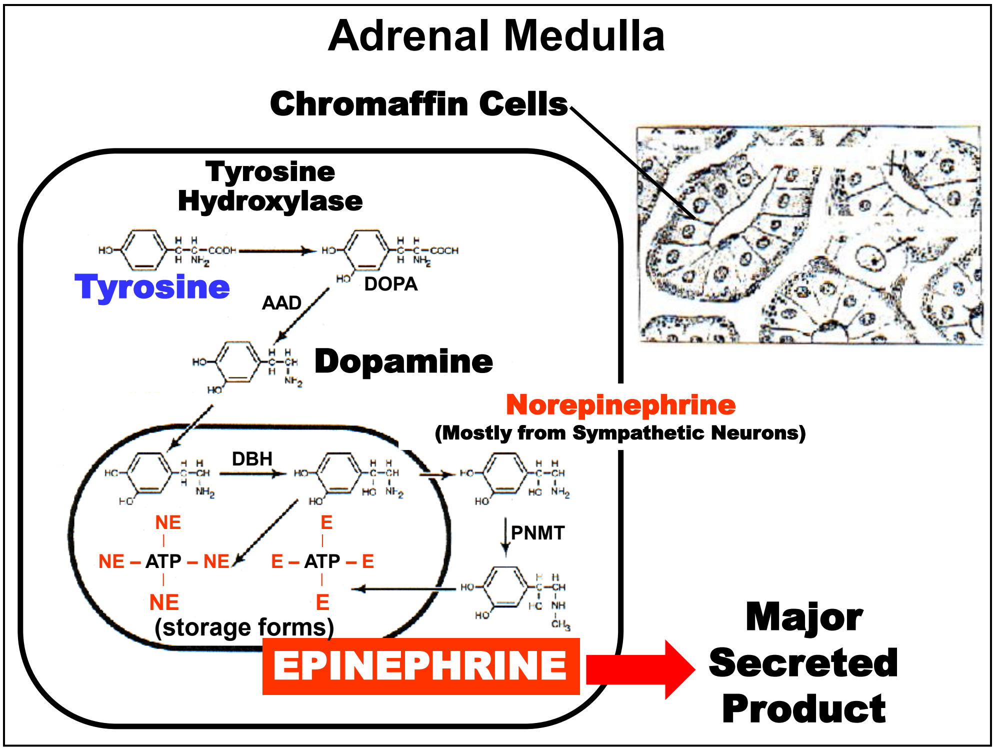
The Adrenal Medulla (*bottom region*) is considered a modified sympathetic ganglion that secretes **Catecholamines** (mainly **Epinephrine**) into blood in response to preganglionic sympathetic stimulation.

Adrenal Medulla Hormone Properties

- **Classification**: **Amino-Acid Derived (Tyrosine)**
- **Biosynthesis**: **Adrenal Medulla**
- **Storage**: **Yes**
- **Secretion**: **Exocytosis**
- **Transport**: **Dissolved**
- **Metabolism**: **Modification** (Short Half-Life: Seconds-Minutes)
- **Mechanism**: **G-Protein Couples Membrane Receptor**
- **Actions**: **Energy Metabolism, Stress Coping**
- **Regulation**: **Autonomic Sympathetic**
- **Disorders**: **Excess: Pheochromocytoma (Neoplasm)**
  Deficiency: Adrenalectomy

Adrenal Medullary Hormone endocrine properties

Highlights are summarized above.



The **Adrenal Medulla** accounts for ≈10% of the adrenal gland mass and is embryologically and physiologically distinct from the cortex (although cortical & medullary hormones often exert complementary actions). The medulla is actually a specialized part of the Sympathetic Nervous System. (*top right*) Secretory cells of the adrenal medulla are called **Chromaffin Cells** and are considered modified **Postganglionic Sympathetic Neurons (Ganglion)** because they arise from neuroectoderm and receive efferent input from sympathetic spinal neurons and secrete catecholamine hormones.

Chromaffin Cells secrete the class of **Amino Acid Derived** hormones called **Catecholamines**, which are derived from the amino acid **Tyrosine**. The most biologically important products are: **Epinephrine, Norepinephrine** & **Dopamine**, which is also the hypothalamic hormone PRIH). Thus, the Major Secreted Product from the adrenal medulla is **Epinephrine** (≈**80%** in humans). The remainder is mostly Norepinephrine & Dopamine. Know this.

Catecholamines: Secretion & Transport

Secretion: Preganglionic Axon **(ACh) = Exocytosis**

Transport:

- **70% Sulfate Conjugated**
- **>> Plasma [Norpinephrine]: Neuron Release**

PLASMA [CATECHOLAMINE]

Catecholamine	At Rest	Activity or Stress
Epinephrine	20-50 pg/ml	150 pg/ml
Norepinephrine	100-400 pg/ml	1000 pg/ml

Affect of Adrenalectomy on [Catecholamines] ???

<u>Secretion</u>. Adrenal catecholamine secretion is initiated by **Acetylcholine** released from the preganglionic axons innervating chromaffin cells. Depolarization of the plasma membrane leads to an influx of Ca^{++} through voltage-gated channels, which triggers **Exocytosis** of cellular storage granules.

<u>Transport</u>. In plasma, ≈70% of norepinephrine & epinephrine are conjugated to Sulfate. These pools may function as <u>Inactive</u> plasma storage forms. The normal plasma levels of unbound <u>active</u> catecholamines are listed in Table 1

Note that At Rest (*middle column*), the plasma concentration of **Norepinephrine is 2-8x *greater* than Epinephrine**. This is because significant amounts of norepinephrine "leak" from sympathetic nerve endings into venous blood. Also notice that the difference increases even more (*right column*) in situations associated with intense sympathetic activity (Activity or Stress). The much greater basal and stimulated norepinephrine levels are due to the large amounts contributed by neuron release.

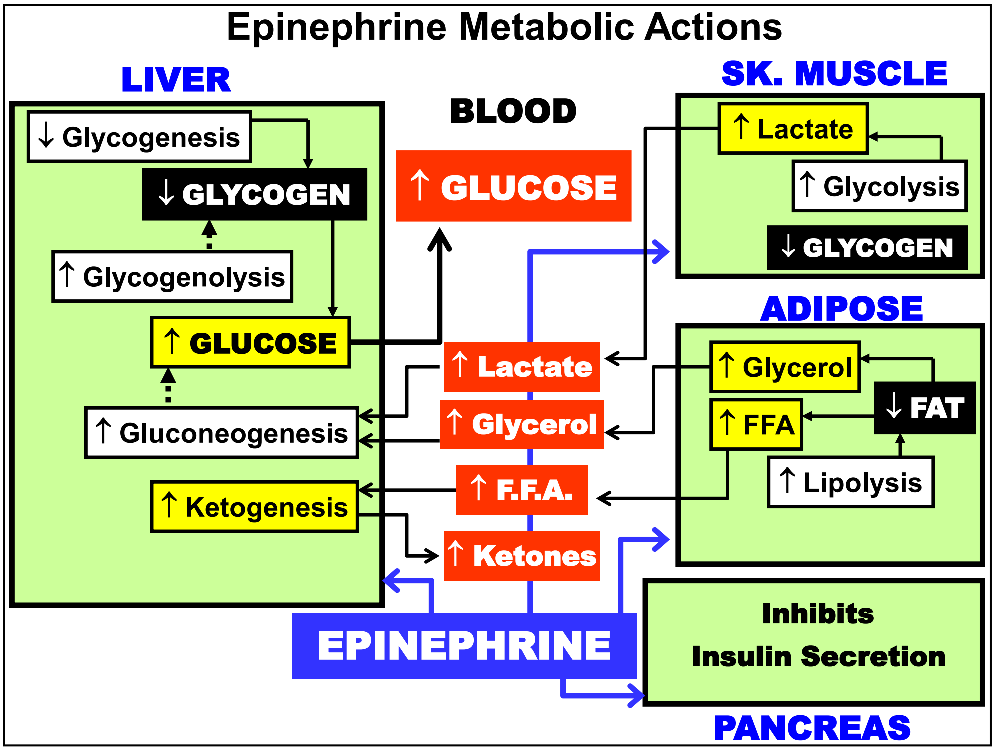
What would be predicted to happen to relative plasma Norepinephrine & Epinephrine concentrations after a complete **Adrenalectomy**?

Catecholamines: Physiological Actions

Target		Responses
Heart		↑ Contraction Frequency & Rate
		↑ Conduction
		↑ Blood Flow (Coronary Dilation)
		↑ Glycogenolysis
Arterioles	•Skin	Vasoconstriction
	•Mucosa	Vasoconstriction
	•Skeletal Muscle	Vasoconstriction or Vasodilation
Fat Metabolism		↑ Lipolysis
		↑ Plasma FFA/Glycerol
Liver Metabolism		↑ Glycogenolysis & Gluconeogenesis
		↑ Blood Glucose
Muscle Metabolism		↑ Glycogenolysis
		↑ Lactate & Pyruvate Release
Bronchial Muscle		Relaxation
GI Tract		↓ Motility
		↑ Sphincter Contraction
Urinary Bladder		↑ or ↓ Sphincter Contraction
Skin		↑ Local Sweating
Salivary Gland		↑ Amylase Secretion
		↑ Watery Secretion
Kidney		↑ Renin Secretion
Skeletal Muscle		↑ Tension Generation
		↑ Muscle Endurance

The adrenal medullary hormones, like the cortical hormones, act on a wide variety of tissues (**Target**-*left column*) to maintain the homeostasis of the internal environment both at rest and in the face of internal and external challenges. Some of the physiological actions (**Responses**-*middle column*) of catecholamines on various tissues are summarized above.

In general, catecholamines enable us to cope with emergencies that have been termed "**Fight or Flight**". Thus, most actions of the medullary hormones mimic those of the autonomic Sympathetic Nervous System. Responsive tissues make <u>NO</u> distinctions between blood-borne catecholamines and those released locally from nerve endings. Note that with only a few exceptions, <u>increased</u> catecholamine secretion acts to <u>increase</u> or enhance the normal functions related to the listed tissues or organ systems.



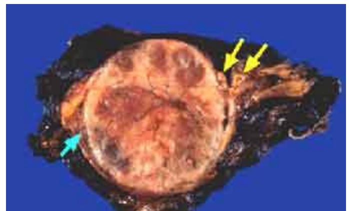
As diagrammed above, catecholamines (**Epinephrine**) have direct effects on Fat (**Adipose Tissue**-*middle right*) and Carbohydrate (**Liver**-*left*; **Skeletal Muscle**-*top right*) metabolism. Indirect metabolic actions are also mediated through effect on Insulin from the **Pancreas** (*bottom right*).

You should know the <u>NET</u> metabolic effect of Epinephrine on **Plasma** concentrations of energy metabolites. You will <u>NOT</u> be required to replicate this flow chart.

However, you should be able to rationalize the dynamics of energy substrate flow between tissues caused by epinephrine and how they result in the changes in blood concentrations above (i.e. **All Increase**).

Catecholamine Disorders

1° HYPOsecretion:
- **Adrenalectomy**
- **Autonomic Compensation**
- **Clinically:**
  - **Hypoglycemia**
  - **Depressed Cardiac Function?**



Pheochromocytoma

1° HYPERsecretion:
- **Medullary Tumor**
- **Pheochromocytoma**
- **Clinically:**
  - **Hyperglycemia**
  - **↑ Heart Contractility**
  - **Hypertension**
  - **Tachycardia**
  - **Anxiety**

HYPOfunction of Adrenal Medulla.

1° **Hyposecretion**. Adrenal medulla insufficiency (e.g. Adrenalectomy) is rare because individuals with otherwise intact autonomic nervous systems are able to largely compensate for any loss in adrenal medullary function. However, individuals developing autonomic dysfunction mainly display Metabolic Impairments such as **Hypoglycemia** (especially with loss of complementary counterregulatory hormones such as Glucagon) and less commonly Depressed Cardiac Functions.

HYPERfunction of Adrenal Medulla: **Pheochromocytoma**.

1° **Hypersecretion**. Medullary Tumor (Pheochromocytoma) arising from chromaffin cells in the sympathetic nervous system is the most common cause, but remain rare (≈2 in one million).

Excessive levels of Epinephrine predictably result in clinical manifestation (*bottom right*) related to its exaggerated actions on glycemic control (Hyperglycemia) and cardiac function (↑ Contractility, Hypertension, Tachycardia, & Anxiety).